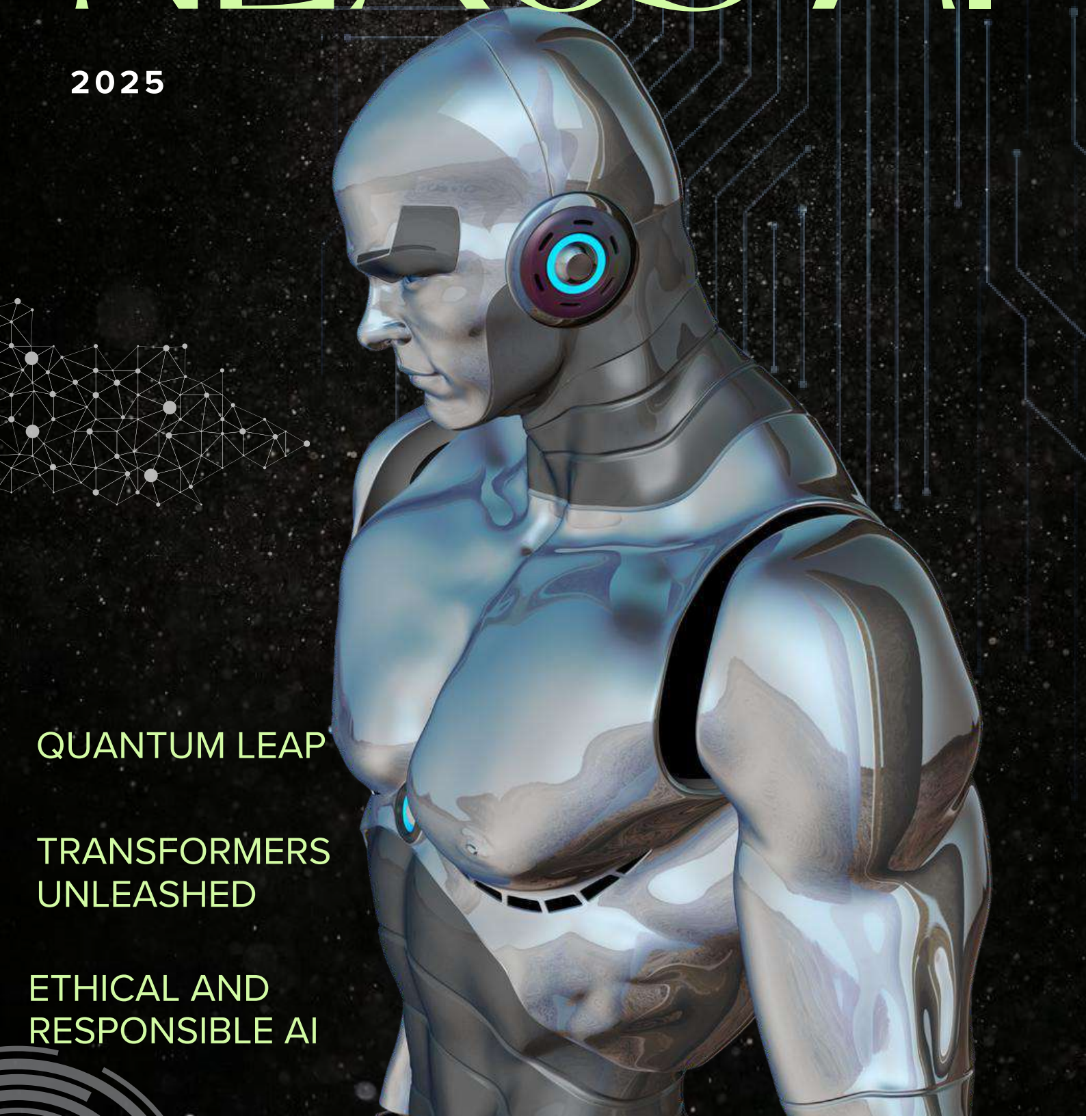
SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

NEXUS AI

2025



QUANTUM LEAP

TRANSFORMERS UNLEASHED

ETHICAL AND RESPONSIBLE AI

SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)
(Established under Section 3 of the UGC Act, 1956) | Re-accredited by NAAC with 'A++' grade | Awarded Category – I by UGC

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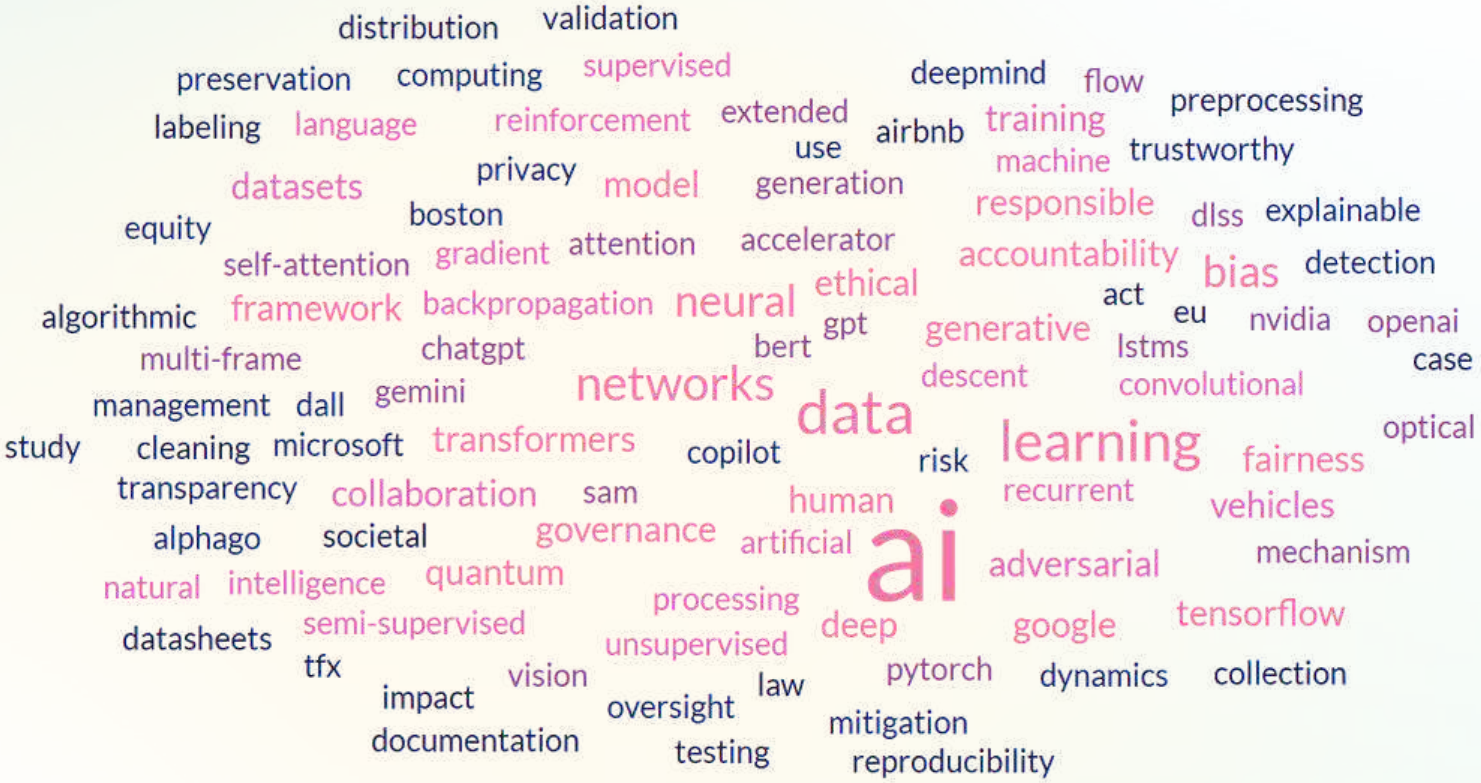
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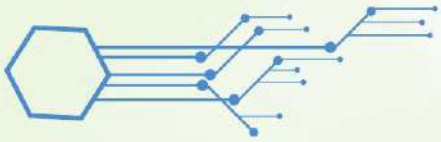
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WORD CLOUD


distribution
validation
preservation
computing
supervised
deepmind
flow
preprocessing
labeling
language
reinforcement
extended
airbnb
training
use
machine
trustworthy
datasets
privacy
model
generation
responsible
dlss
explainable
equity
boston
self-attention
gradient
attention
accelerator
accountability
bias
detection
algorithmic
framework
backpropagation
neural
ethical
act
eu
nvidia
openai
multi-frame
chatgpt
bert
gpt
generative
lstms
case
management
dall
gemini
networks
descent
convolutional
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cleaning
microsoft
transformers
copilot
data
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transparency
collaboration
sam
risk
fairness
alphago
societal
governance
human
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vehicles
natural
intelligence
quantum
artificial
ai
adversarial
mechanism
datasheets
semi-supervised
processing
deep
google
tensorflow
tfx
unsupervised
vision
pytorch
dynamics
collection
impact
law
oversight
mitigation
documentation
testing
reproducibility

VISION & MISSION



Vision

To evolve as a centre of excellence in Artificial Intelligence and Machine Learning by fostering quality education, research, and industry collaboration to develop globally competent professionals contributing to societal enrichment.

Mission

1. To empower students with comprehensive foundations in Artificial Intelligence and Machine Learning, by enabling technical, analytical, and leadership skills to excel in multidisciplinary domains.
2. To promote innovation and applied research through collaborative partnerships with academia, industry, and government bodies.
3. To nurture ethically responsible AI professionals committed to lifelong learning and the development of sustainable and inclusive intelligent solutions.
4. To develop global competence through interdisciplinary learning and active collaboration with international research and industry partners.

MESSAGE FROM DIRECTORS, DESK



Dr Ketan Kotecha

Dr Ketan Kotecha
Director,
Symbiosis Institute of Technology, Pune.

It gives me great pride and joy to present this edition of the AI Magazine, an academic platform that captures the spirit of innovation, inquiry, and excellence driving our Artificial Intelligence and Machine learning community.

Artificial Intelligence is not merely a technology it is a transformative discipline that challenges us to think critically, act ethically, and innovate responsibly. As we witness AI revolutionizing every domain of human activity, from education and healthcare to sustainability and governance, it becomes our academic duty to prepare the next generation of thinkers and leaders who will shape this transformation.

At Symbiosis Institute of Technology, Department of AI & ML we are deeply committed to impart quality education with research insights for developing competent global engineers in the field of Artificial Intelligence and Machine Learning to solve societal problems nurturing human-centered AI education where technical expertise is complemented by ethical awareness, creativity, and interdisciplinary collaboration. Our students and researchers exemplify this spirit through their projects, research endeavours, and contributions that push the boundaries of AI while addressing real-world challenges.

This magazine is a testament to their enthusiasm and perseverance. It showcases not only their technical achievements but also their ability to think critically about the role of AI in society. I commend the editorial team, contributors, and mentors who have made this initiative possible through their passion and dedication.

May this issue inspire all readers to explore, question, and innovate to see AI not just as an academic pursuit, but as a means to build a future that is intelligent, inclusive, and responsible.

MESSAGE FROM THE HOD, DEPARTMENT OF AI-ML



Dr Sumit Kumar

Dr Sumit Kumar
HOD, AI & ML Department
Symbiosis Institute of Technology, Pune.

It gives me great pleasure to present the first edition of our Department's Technical Magazine, centered around the fascinating frontiers of Quantum Computing, Transformers, and Reinforcement Learning — three pillars that are redefining the future of intelligent systems.

Quantum computing promises to revolutionize computational capabilities, enabling machines to solve problems once deemed intractable. Transformers have transformed the landscape of natural language processing and vision, empowering machines to understand and generate human-like data with unprecedented precision. Reinforcement learning, inspired by the principles of trial and feedback, continues to drive breakthroughs in autonomous systems and decision-making intelligence.

As these domains converge, they represent not just technological innovation but a glimpse into the next era of human–machine collaboration. I am proud to see our students and faculty engaging with these cutting-edge concepts, exploring their applications through research, projects, and creative expression.

This magazine stands as a testament to the intellectual curiosity and technical excellence of our department. I congratulate the editorial team and contributors for curating a thought-provoking collection of insights and innovations, and I hope this issue inspires readers to continue pushing the boundaries of what is possible.

MESSAGE FROM THE FACULTY EDITOR



Dr. Dhanashree K. Barbole

It It gives me enormous pleasure to introduce this issue of the Annual Magazine of Department of Artificial Intelligence & Machine Learning, highlighting the wisdom, creativity, accomplishments, and experiences of our students and faculties.

AI & ML are changing the world through innovation and technology. This magazine is an example of our department's dedication to learning, research, creativity, and innovative development.

It is my sincere pleasure to thank our students, faculties, and editorial team members for their efforts that have made it possible to compile this issue of ideas, accomplishments, and memories.

Let this magazine serve as an inspiration for everyone to learn, explore, innovate, and make contributions to the dynamic world of Artificial Intelligence and Machine Learning.

Happy Reading!

Dr Dhanashree K Barbole
Assistant Professor
AI ML Department
Symbiosis Institute Of Technology

MESSAGE FROM THE STUDENT EDITORS



Riddhi Poddar

It has been a rewarding experience serving as the editor for our Artificial Intelligence and Machine Learning Department Magazine. Over the past few weeks, I had the opportunity to review a wide range of submissions from passionate and innovative minds across our department. Every article reflected genuine curiosity, creative thinking, and a strong drive toward exploring emerging technologies in AI. I appreciate the hard work contributed by all participants, and I encourage everyone to keep exploring, questioning, and creating. Thank you to our faculty members and peers who supported us throughout the process. I hope this magazine inspires more conversations and innovations in the world of AI.

Parth Koshti

Serving as the Editor of our Artificial Intelligence and Machine Learning Department Magazine has been a greatly fulfilling experience. Reading through the articles and work of our peers has opened my eyes to the creativity, thoughtfulness and commitment that exists in our department. Each article provides a different perspective on the ways that AI is influencing the world around us. I am grateful to all of the contributors for their time, as well as to our faculty members for their ongoing support and guidance. I hope this magazine inspire every student to be inquisitive, think outside the box, and continue to explore the endless possibilities of AI.



MEET OUR TEAM

STUDENT



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Parth Koshti
Editor



Dharna Sharma
Proofreader



Deshna Tendulkar
Design and Layout Editor

MEET OUR TEAM

FACULTY



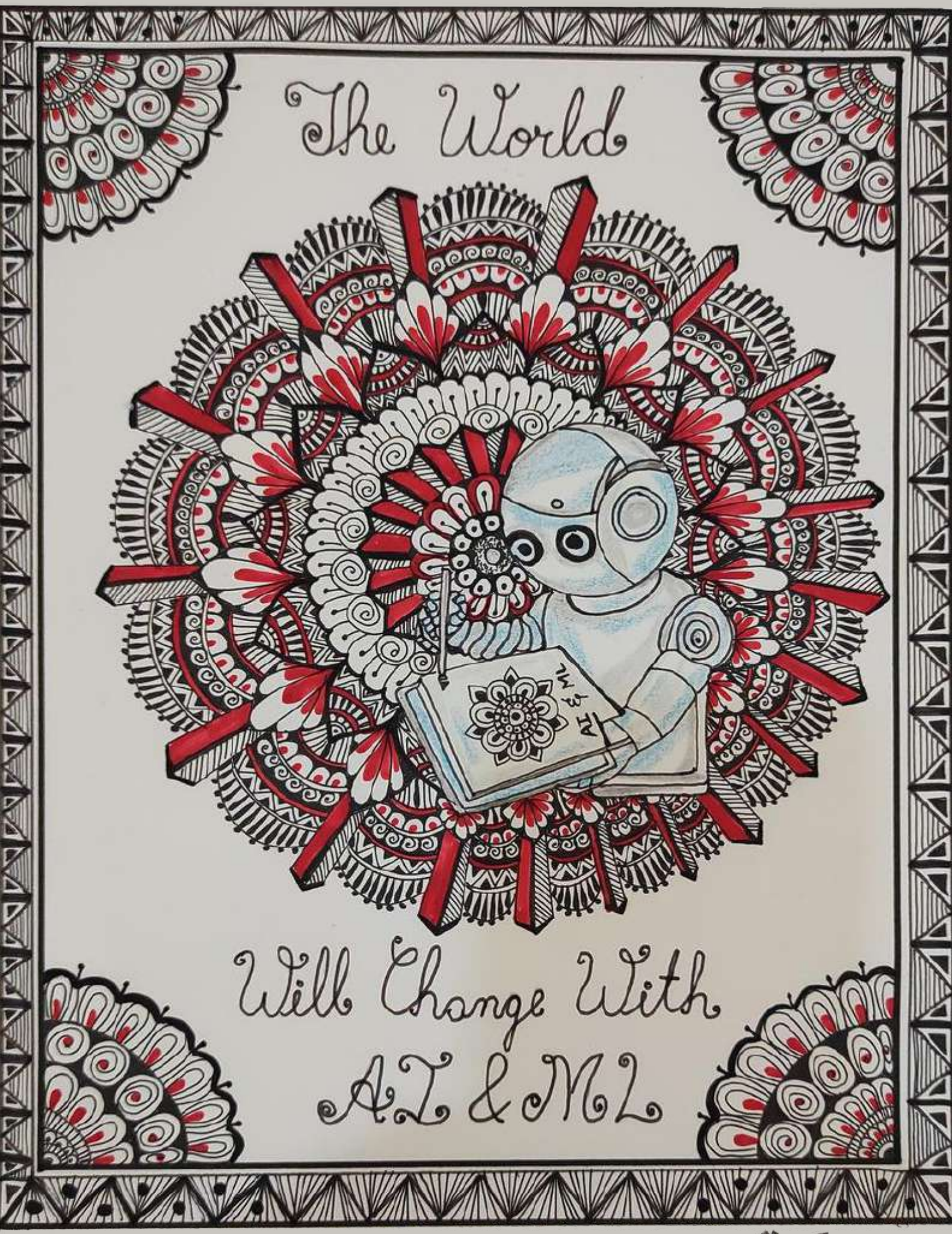
Dr. Dhanashree K. Barbole
Magazine In-charge



Prof. Ashwini Tande
Co-Head



Dr. Pooja Kamat
Co-Head


The World
AI & ML
Will Change With
AI & ML
Suman

AI: From Logic to Imagination

"AI: From Logic to Imagination"
It began with Machine Learning's dawn,
Where data met decisions head-on.
Regression lines and SVMs so wise,
Taught systems to see with human eyes.
From spam detection to credit score,
Predictive models opened the door.
Yet shallow layers could not feel —
Complex truths remained concealed.
Then Deep Learning rose to take domain,
With neural layers trained through loss and gain.
Convolutional nets parsed image space,
Extracting features with learned grace.
RNNs and LSTMs found their way,
Through time and text, where meanings play.
Now cars drive straight through city streams,
And AI reads our spoken dreams.
But the story turned another page,
When Transformers took the stage.
Attention whispered, "Focus here,"
And GPTs began to steer.
Now Generative AI paints and sings,
Creates new worlds with latent strings.
In labs and art, in code and rhyme,
It co-authors future's paradigm.
From chatbots that reason to robots that feel,
To drug design and vision unreal,
AI transcends — both tool and friend,
Where data and dreams now blend.

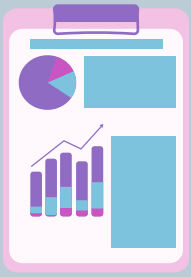


-Dr. Supriya Mahadevkar
Assistant Professor,
AIML Department

AI ETHICS

Building a Fair & Trustworshy Future





Datasheets for Datasets: A Review

Raam Bhanushali, AIML 2024-28

Introduction and Motivation

Working on machine learning projects has taught me that not every dataset is a good quality one. It happens almost all the time - you download a dataset, train a model, get very good results and just after that you find out that your data was collected in such particular condition it makes any sense for whatever real world application. It’s infuriating and highlights one of the many problems in AI: the black-box nature and poor documentation of our data on which we work.
This is the motivation behind the concept of "Datasheets for Datasets". I have now reviewed the framework proposed by the authors to solve the problem and think it should be a standard for every data creator and data user in machine learning.

The Core Idea:

Wherever you go, you see electronic components and their corresponding data sheets. They explain the component's specifications and operating limits, as well as proper usage, and provide you a guide. The "Datasheets for Datasets" proposal applies the same idea to machine learning (ML) datasets. The aim is to create a uniform document answering all the key questions on a dataset's history, what it is made of, and how the dataset can be utilized.

There is also the aim to:

- Foster Transparency: It’s a mystery how some datasets are constructed, and a datasheet compels the builder to record their trail.
- Combat Bias: Models, to a degree, amplify the societal bias embedded within the data they are trained on. Early on, we can capture potential bias issues by documenting the data composition and collection.
- Enhance Reproducibility: Other researchers will be able to focus on the core dataset and aligned datasets, documenting their process and results for the usefulness of others.
- Eliminate Misuse: It prevents problematic datasets, like a dataset of celebrity faces, from being used inappropriately in a sensitive facial recognition security system.

The Framework:

The suggested datasheet is not a sketch. It is a full document divided into seven coherent sections, which corresponds to a dataset's lifecycle. More important ones, I would explain them as follows:

1. Motivation: This asks why the dataset is created. Understanding the purpose of the creator and the funder is important in knowing possible biases.

2. Composition: This is the dataset's name. What is the type of data: images, text? How many data points? What are the labels? It is very important that it asks about the presence of sensitive data— race, gender— errors, and duplicates.

3. Collection Process: This is the "how." Was data scraped from the web, collected via sensors, or manually labeled? Were the people who collected or labeled the data fairly compensated? This part is important for evaluating the quality and the ethical aspects surrounding the data.

4. Preprocessing/Cleaning/Labeling: Almost never do we use completely raw data. This part outlines all the steps taken to clean, filter, and label the data. This is super important because these steps can introduce their own biases.

5. Uses: This may be the most pertinent section for students. It outlines previous uses and, most importantly, specifies tasks the dataset is not appropriate for. This directly helps to mitigate misuse.

6. Distribution: How can you access the data? What are the terms for its use? This may seem trivial, but unclear licensing can derail practical work for a project.

7. Maintenance: Is the dataset currently being maintained or updated? Who should be contacted with any inquiries? This indicates the dataset should be a living document, not an archive.

The Real World Impact and The Challenges:

This isn't just a theoretical concept. Large technology firms, such as Google and Microsoft, have implemented internal versions and have inspired comparable tools called "model cards." This indicates that the industry understands the need for responsibility. However, implementing this widely isn't without challenges:

- It Takes Work: Creating a thorough datasheet requires significant time and effort from creators, which can be a hard sell without proper incentives.
- Dynamic Data is Tricky: For datasets that are constantly updated (like a live social media feed), keeping the datasheet current is a major challenge.
- It's Not a Magic Bullet: A datasheet can't catch every single bias or foresee every possible misuse. It's a tool for better judgment, not a substitute for it.

Government Rules: How AI Just Got Real

Priyansh Johri, AIML 2024-28



Hello everyone, In our discussions regarding AI ethics, especially concerning issues like algorithmic bias and fairness, I used to think it was something we would have to dodge for a few decades. It felt important, but I thought we would have time before we had to worry about it. Now, it really does feel like that time has come. Tech companies are not the only ones trying to dictate the future. Governments are stepping in and we all know how that changes the game.

Responsible AI started out with a set of guiding principles and for a long time that was the only way to approach it. It was a good first step, but in all honesty we all know that it was just a first step, and that there was a complete lack of enforcement in the adherence to those set principles.

Europe Just Changed the Game

In March 2024, the European Union officially passed the AI Act. This isn't a suggestion it's a law and a pretty strict one. Think of it like a food safety inspector, but for artificial intelligence. The law sorts AI into different risk levels.

- **Unacceptable Risk:** Some AI is just straight-up banned. Things like government "social scoring" systems that rate citizens' behavior are illegal. So is using AI to exploit people's vulnerabilities.
- **High Risk:** This is the big one. AI used in hiring, banking loans, and critical infrastructure like water and energy falls here. These systems now have to follow a long list of rules before they can be used. They must be transparent, accurate, and overseen by humans.
- **Limited Risk:** This includes things like chatbots. The main rule here is simple: you have to tell people they are talking to an AI. If a European company plans to implement AI technology to scan resumes, the AI must demonstrate non-discriminatory capabilities. If the company fails to comply, it will incur hefty penalties. Likewise, being accountable is no longer a favorable concept it has become a mandatory requirement by law.

What About the United States?

The U.S. is doing its own thing, but it's just as serious. In October 2023, President Biden signed a major Executive Order on AI.

It's like a giant to-do list for the entire U.S. government. The order tells federal agencies to:

- **Create Safety Tests**: The National Institute of Standards and Technology (NIST) is setting up an AI Safety Institute to create standard tests for the most powerful AI models, like red-team exercises to find their weaknesses.
- **Fight Deepfakes:** It pushes for developing digital "watermarks" for AI-generated content. This means a video, image, or audio clip made by the government would have a label, so you know it's AI. This is a direct response to the fear of fake news and disinformation.
- **Protect Privacy**: It calls for stronger privacy guidelines to protect our data from being misused by AI.

What Does This Mean for Us?

Transitioning from discussing the ethics of AI to putting them into practice is something to consider seriously. This shift indicates that stakeholders in development are being made responsible. Companies are no longer in the position to just say, "Our AI is fair trust us." They have to back that claim with evidence. This is important for us as students concepts like bias, transparency, and accountability aren't something that will come into play in the future. They are relevant to us right now. AI ethics have moved beyond being only applicable to technology-related courses. They are now also part of the law, politics, and public policy. Drafting legislation is only the first step and understanding the legislation will help in planning the future of AI.





How the World is Actually Using the NIST AI Framework

Deshna Tendulkar, AIML 2024-28

Let's be truthful who hasn't used ChatGPT, experimented with DALL-E to create some interesting art, or seen those amazing Boston Dynamics robots doing backflips, They are really amazing, aren't they because Artificial Intelligence has moved from a sci-fi concept to something that is really influencing our daily lives. It is so rapidly evolving that it is honestly a little frightening. There is virtually no end to what AI can do now, from self-driving cars, and medical AI that assists with diagnostics, to tools that can write your essays and produce incredible art. With huge power comes huge responsibility. AI technology involves risks that, in some cases, could be serious risks to people, businesses, and the world. An AI that makes biased decisions, gives inaccurate medical diagnoses, or excessively consumes energy harming the environment, are some scenarios that could have serious consequences. That's why the National Institute of Standards and Technology (NIST) the group that sets technological guidelines for the US has produced the AI Risk Management Framework (AI RMF 1.0) document. It is quite a lengthy text. No need to worry, I have read it and can explain the implications for us.

Real-World Application 1: Governing a Hospital's Diagnostic AI

- **GOVERN:** Before anything else, the hospital forms an oversight committee with doctors, data scientists, lawyers, and even patient advocates. They create a strict policy: the AI is a "second opinion" tool, and a human radiologist must always make the final diagnosis. This establishes accountability from day one.
- **MAP:** The team maps out the risks. What if the AI was trained mostly on data from one demographic and performs poorly on patients from another? What happens if the system goes offline during a critical diagnosis? They identify these context-specific risks upfront.
- **MEASURE:** They don't just trust the AI company's marketing. They rigorously test the model on their own, diverse set of patient data. They measure not just accuracy, but fairness—ensuring it works equally well for all patient groups. They also test for robustness, seeing if slightly blurry or off-angle X-rays confuse the system.
- **MANAGE:** Based on their measurements, they manage the risks. They might decide to only use the AI for a specific type of scan where it performs best. They set up a clear incident response plan: if a radiologist disagrees with the AI, it flags the case for a second human review and the data is used to retrain and improve the model.

Real-World Application 2: Managing Bias in a Bank's Loan Application System

- GOVERN: The bank's leadership mandates that "fairness" is a non-negotiable requirement, on par with profitability. They appoint a senior executive as directly accountable for the AI's compliance with fair lending laws.

- **MAP:** The team works with sociologists and ethics experts to understand the potential for harm. They identify that the biggest risk is the training data itself, which contains decades of historical loan decisions that may reflect past biases.
- **MEASURE:** Instead of just looking at overall approval rates, they constantly slice and measure the AI's decisions. They check approval rates and interest rates by zip code, race, and gender to detect any disparate impact. They use specific metrics for fairness to quantify any potential discrimination.
- **MANAGE:** When the measurements find a potential bias, they don't ignore it. They manage it by retraining the model with de-biasing techniques, removing problematic features from the input data (like using zip code as a direct proxy for income), and creating a clear appeals process for applicants who believe they were unfairly assessed by the AI.

Real-World Application 3: Ensuring Safety in a Self-Driving Car Company

- **GOVERN:** The company builds a culture where any engineer can "red flag" a safety concern without fear. Safety reviews are integrated into every single step of the development cycle.
- **MAP:** They conduct "what-if" scenarios for everything. What if a sensor fails in heavy rain? What if the car encounters a construction zone not in its maps? This helps them identify thousands of potential edge cases and risks.
- **MEASURE:** They go far beyond standard software testing. They run millions of miles in simulation, testing the AI's performance in these risky edge cases. They physically test the cars in controlled environments to measure its reliability and resilience against unexpected events.
- **MANAGE:** The primary risk management strategy is a "defense in depth" approach. The car has multiple, redundant systems (cameras, lidar, radar) so if one fails, others can compensate. There is also a clear protocol for the AI to safely disengage and request human intervention when it encounters a scenario it cannot confidently handle.

The Human Element: Still the MVP

One of the coolest things about this framework is how much it puts humans front and center. It is not just about fancy algorithms or lines of code it is about us having diverse teams, human oversight, and clear responsibilities is key. People bring context, ethics, and the ability to question and double check what AI spits out. We are not just passengers here, we are co-pilots, constantly steering and fixing the course.

What is the Bottom Line?

The NIST AI Risk Management Framework should not be understood as a set of binding regulations. Instead, it is a framework that, case by case, can adapt as AI technology rapidly evolves. It is designed to encourage rational consideration of powerful yet safe, equitable, and beneficial AI systems by corporations, public authorities, prospective AI developers, and us. This is a vital advancement to guarantee that the increase of AI in our everyday lives is beneficial, not detrimental. Imagine the most complex game in the world. It’s a matter of life and death, so you need a detailed instruction manual.

TRANSFORMERS

(BERT)

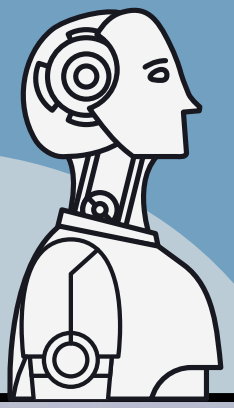
Language Model Architecture



DLSS 4: Fake Frames, Real Tech

A technical breakdown of Nvidia's DLSS 4 technology and multi-frame generation.

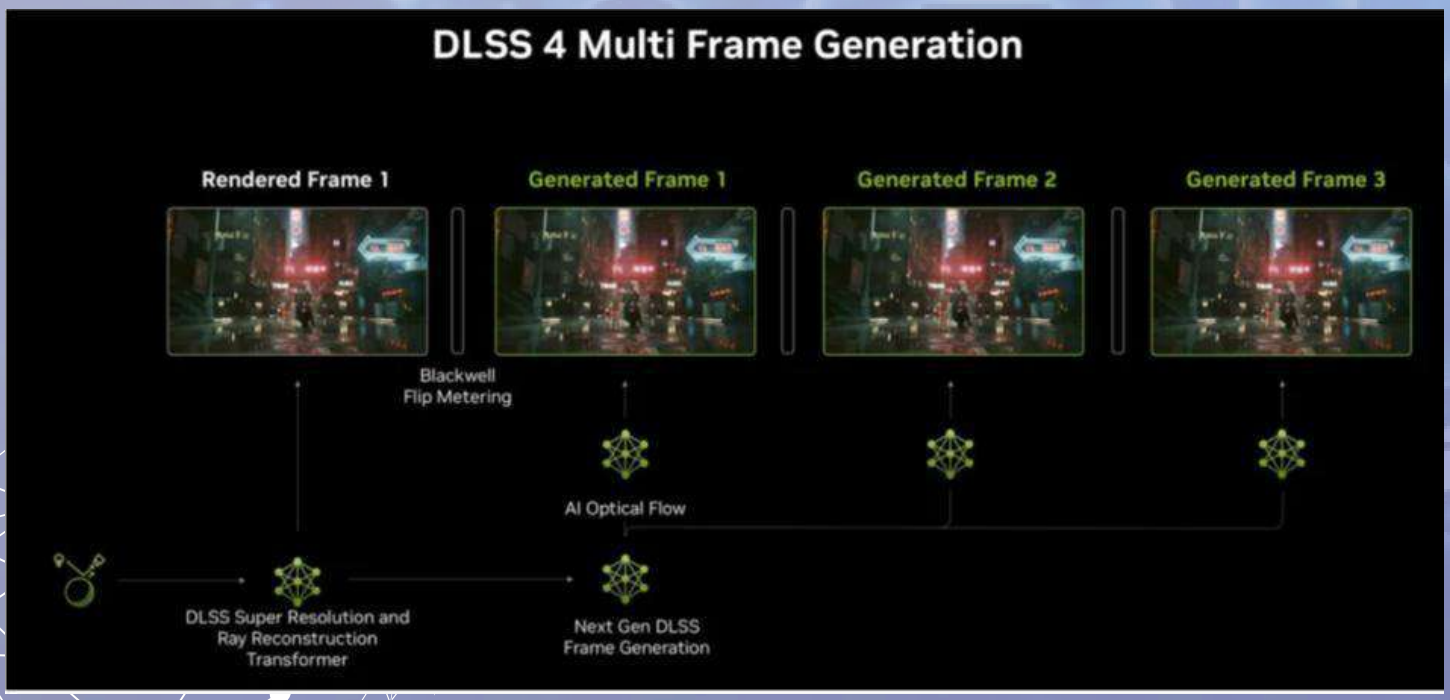
Shantanu Shaji, AIML 2024-28



"If your GPU didn't draw it, is it even really performance?" That's the kind of question lighting up every comment section since they released Nvidia's Multi-Frame Generation technology along with their RTX 50 series cards. Quadruple the frames and at 4k resolution? Sounds fake, until you look closer.

Nvidia achieves this using DLSS 4 (Deep Learning Super Sampling), powered by a Transformer based AI model that doesn't simply upscale frames, it generates. Making use of temporal data (motion vectors, depth maps, frame history), DLSS predicts what the scene should evolve. It makes use of their Optical Flow Accelerator to track fine object movements and interpolate new frames between the real ones. With this, Nvidia is able to generate 4 times more frames than usual.

Unlike the previous models which worked on convolutional neural networks (CNNs), DLSS 4 uses a Transformer based model, the same architecture that powers LLMs like ChatGPT. This allows it to evaluate the whole frame along with its physics data, and not just a small local region. Along with the data from multiple past frames and motion data, the model understands how objects move and how the camera moves, allowing it to create frames that feel coherent and not just similar.



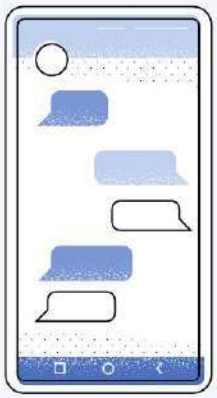
The big development with DLSS 4 is its Multi-Frame Generation (MFG) system, where the game engine actually renders only one out of every four frames, while AI generates the rest. This is made possible by the upgraded Optical Flow Accelerator system, which is what helps understand the flow between two actual frames, helping the MFG to create accurate frames.

But of course, the people's concerns aren't baseless and the tech isn't flawless. There is visible artifacting that might create a slightly jarring experience, and thinner objects like wires or small particles can cause visual glitches. And while DLSS can create a "higher frame rate", it comes at the cost of increased latency and less responsiveness.

It is not "free" performance, and certainly not as beautiful as native rendering, but DLSS 4 shows that smarter rendering might be the future. For some the trade off between responsiveness and visual fidelity might be worth it. For others, nothing beats native frame rendering. As the line between real and generated continues to blur, it is up to you to decide what kind of performance really matters to you.

How Transformers Are Powering ChatGPT, Google Search, and GenAI Tools Today

Sejal Dubey, AIML 2022-26



Ever wondered how tools like ChatGPT, Gemini, and other AI assistants work behind the scenes? From helping you write assignments to generating software code, these tools are changing the way we study, work, and even think. But what powers them under the hood? The answer lies in a breakthrough deep learning model called the Transformer. Let's unlock how this technology works and why it's such a big deal.

What Is a Transformer?

The Transformer architecture was introduced in 2017 in a research paper titled "Attention is All You Need". It marked a turning point in AI.Technically, a transformer is a type of deep learning model that helps machines understand relationships between words ,not just in order, but in context. Unlike earlier models that read sentences word by word, transformers can analyze the entire sentence at once using a technique called self-attention. This allows them to figure out which words are most important, even if they're far apart.They also work in parallel, meaning they can process information much faster and more efficiently , which is essential when training on massive datasets!

Key Milestones: BERT, GPT, and Beyond

The first big leap came with BERT (Bidirectional Encoder Representations from Transformers), which Google introduced in 2018. BERT improved how Google Search understands complex questions — like "Can I pick up medicine for someone else?" by focusing on the meaning of the entire sentence rather than individual keywords.Then came GPT (Generative Pre-trained Transformer), which not only understands text but can also generate it. This is what powers tools like ChatGPT, enabling them to write essays, explain concepts, generate code, and even carry on meaningful conversations.

Other transformer-based models include T5 for translation and summarization,ViT (Vision Transformers) for image recognition and SAM (Segment Anything Model) by Meta, which can detect objects in any image instantly. Transformers in the Real World Google Search uses transformers to understand what you're really asking. ChatGPT and Gemini use them to generate smart, human-like responses.Microsoft Copilot uses GPT models to help users write emails, summarize documents, and automate daily work. Amazon Alexa uses them to better interpret voice commands.YouTube uses them to recommend videos based on your preferences.Even in fields like healthcare, finance, and education, transformer-based models are used for summarizing reports, analyzing legal documents, and building intelligent tutoring systems. The transformer isn't just a research paper anymore , it's the foundation of modern AI. Understanding it helps us appreciate how far technology has come, and where it's heading. Whether you're a casual user or an aspiring developer, understanding transformers gives you a glimpse into the future of intelligent systems , and a chance to be part of building it.

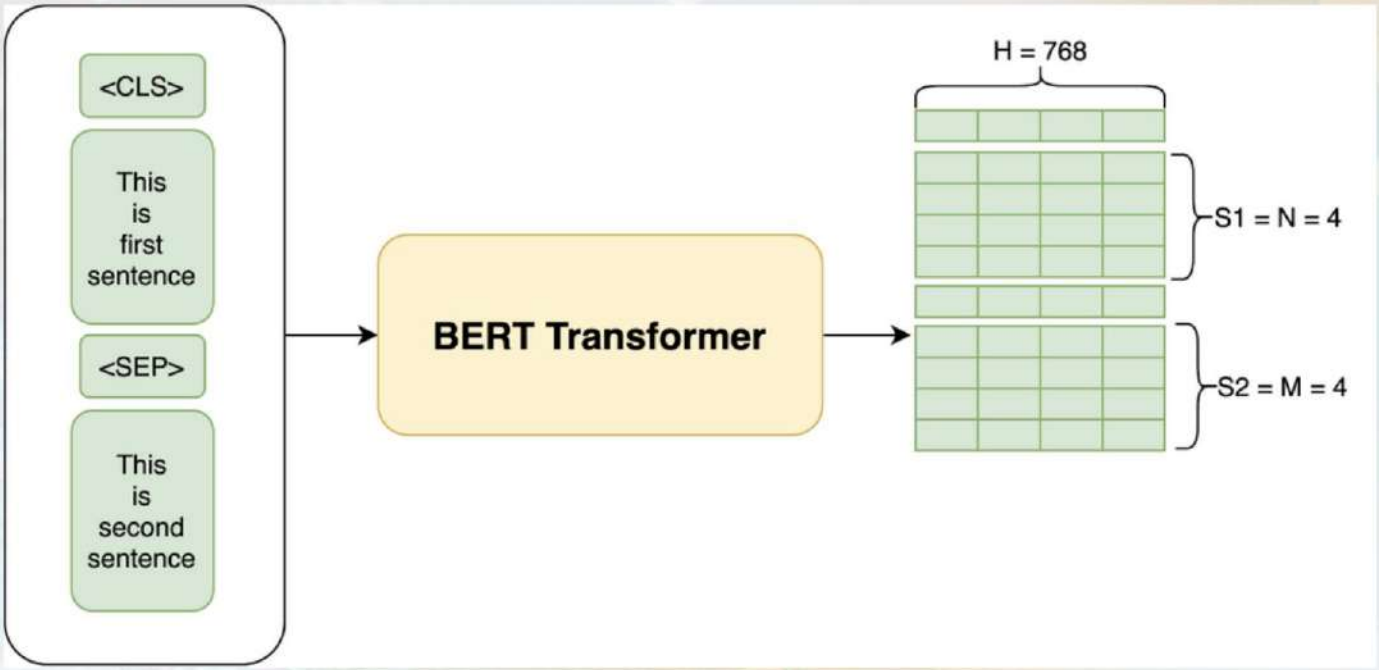
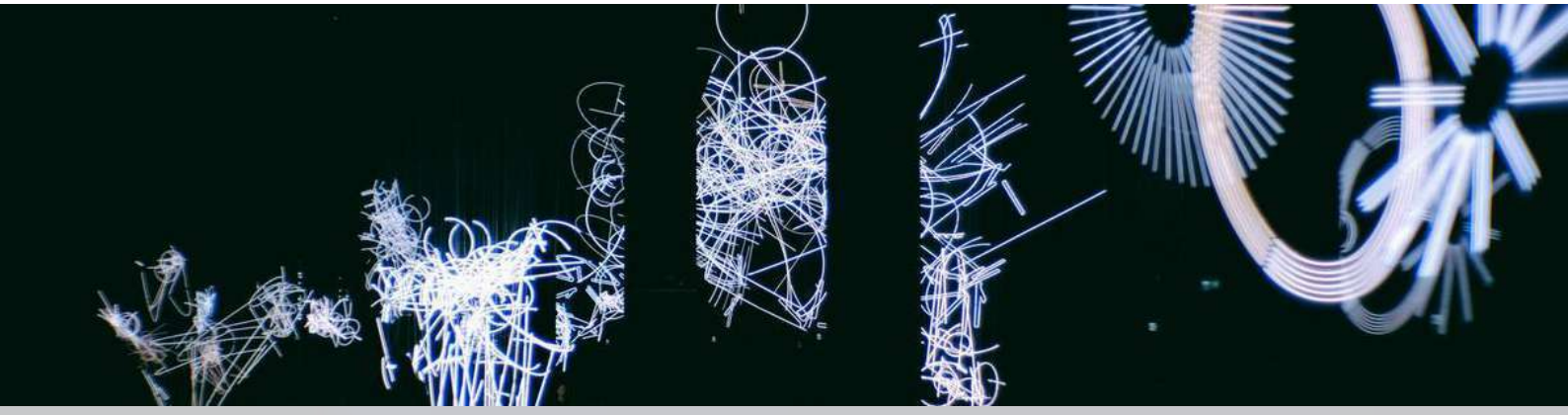




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REINFORCEMENT LEARNING





Learning by Trial and Error: An Interesting Case for Reinforcement Learning

Riya Shukla, AIML 2022-26

We live in a world where automated machines recommend to us our next Netflix Binge, drive cars, play games like chess on our behalf, but have we ever thought about how all this happens? Here comes a quiet but revolutionary branch of AI that powers much of this intelligence: Reinforcement Learning (RL). Think of it as the art of learning from experience. But wait … but what is machine learning used for then? Isn't ML doing all of this? What makes RL so special?

Let's Walkthrough the Learning Loop

Reinforcement Learning is like training a baby—except the baby is an algorithm and the treats are numeric rewards.
How RL exactly works?

- An agent (like a robot or algorithm)
- In an environment (like a game, maze, or financial market)
- Takes actions
- Receives rewards or penalties
- Learns from feedback and aims to maximize total rewards over time

This feedback loop is the main crux of RL. Unlike supervised learning, where the answers (labels) are known beforehand, RL thrives in ambiguity and exploration.

Didn't this sound so much like Human?

What sets RL apart is how natural it feels. As babies, we touch, taste, fall, and get up again. We learn what's good, bad, or boring based on consequences. That's reward-based learning—and RL mimics this beautifully.
What sets RL apart, let's think:
In supervised learning, a teacher tells/guides us towards the right answer. In reinforcement learning, the world tells you when you're wrong by actually interacting with the environment.

Real-Life Adventures of RL

There is something amazing about RL, we all have experienced it without realising we are actually using it in our day to day life.

Let's try to relate:

- AlphaGo beating world champions – It taught itself strategies nobody had ever played before.
- Autonomous cars (Tesla) – Navigating traffic lights, pedestrians, and road rage, all through trial-and-error.
- Robotic arms – Learning to grasp fragile objects, adaptively.
- Financial trading agents – Making split-second investment decisions to maximize returns.

Let's Dive into the Core Algorithms RL

At the core of RL lie a few key algorithms:

- Q-Learning: An off-policy method that learns the value of action pairs
- Policy Gradient Methods: Directly tweak the agent's strategy
- Actor-Critic Models: The actor decides what to do; the critic evaluates it
- Deep Q-Networks (DQN): Combine RL with deep learning to learn directly from pixels

And now, there's buzz around Multi-Agent RL, Offline RL, and Meta-RL—where agents teach each other, learn from past experiences, or learn how to learn.

What Makes a Good Reward?

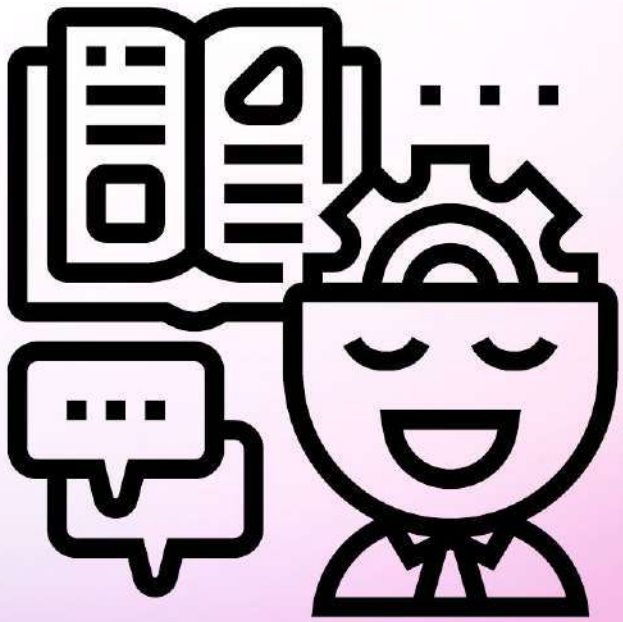
Designing the reward function is like writing a poem with no words. Too simple, and the agent cheats. Too complex, and it never learns.
Want your robot to learn to walk? Reward it for staying upright and moving forward—not just moving its legs. That's the art of reward shaping.

The Philosophy Behind RL

At its core, Reinforcement Learning isn't just a tool—it's a philosophy of intelligence. It embraces uncertainty, celebrates mistakes, and thrives on adaptation.
In a sense, RL isn't just how machines learn. It's how life learns.

Final Takeaway:

In a world where everyone is obsessed with immediate rewards and labelled data, Reinforcement Learning whispers a different truth:
"Sometimes, the only way to learn is to try… and fail."
And maybe, just maybe—that's the most human way to learn of all.



EXPLAINABLE AI (XAI)

Unveiving the Black Box



Why Your "Fair" AI Might Be Totally Missing the Point



Pratham Budhauliya, AIML 2024-28

Hey everyone so in our AIML branch, we're always learning how to build smarter algorithms and more efficient systems. We're basically taught that any problem can be solved with the right code and enough data. But what happens when we try to solve something as messy and human as fairness with clean, logical code?

Turns out, it's way more complicated than it looks. I just read this mind-blowing paper that completely changed how I think about "fair AI." The researchers found that when we try to make AI fair, we often make these huge mistakes because we're treating human problems like math problems. They call these mistakes "abstraction errors" - basically, we're oversimplifying reality to make it fit into our code.

Trap 1: The Framing Problem - Seeing Only the Code

In our class, one of the concepts we learn is how to decompose a problem. But when we do this with fairness, we tend to focus purely on the algorithm the inputs, the code, the outputs, the results and nothing else. Take a video game for instance is it fair to only analyze the code and completely ignore how the countless players interact with the game, yet this is the situation with most of the “fair AI” systems. Take risk assessment tools used in courts for instance. An algorithm can be mathematically fair, but if biased judges are using it, does the fairness really count as the code cannot be compartmentalized. It has to be looked at in conjunction with how the people interact with it in the world.



Trap 2: The Portability Myth - One Size Doesn't Fit All

In AIML, we enjoy reusable code. "Write once, use everywhere!" But fairness doesn't work like that. Take a "fair" hiring algorithm from a tech company and use it for college admissions. The context is completely different! What makes hiring fair might be totally different from what makes admissions fair. The paper calls this the portability trap- we can't just copy-paste fairness from one situation to another.

Trap 3: The Formalism Trap - Confusing Metrics with Meaning

This one resonates personally for me. As optimization specialists, we enjoy simplifying complex targets to loss functions. The fair-ML community has come up with several mathematical definitions for fairness: demographic parity, equalized odds, individual fairness, and more.

Nonetheless, these are the realities we cannot ignore: equity is not a primitive value. It is procedural (how the decision is made), contextual (relative to the given situation), and skeptical (disagreements about fairness exist). This was perfectly illustrated in the COMPAS controversy - disparate mathematical definitions yielded opposing conclusions about the same model. Essentially, we are trying to optimize a socio-coordination problem.

Trap 4: The Ripple Effect - Ignoring Model Deployment Dynamics

In machine learning research, we often conclude our evaluation with the performance on the test

set. However, models in real world scenarios have a much larger impact on changing user behavior in ways that we cannot foresee. Take the example of a "fair" credit scoring model. If automated systems bias lending decisions, or if applicants exploit the scoring model, or if the credit model continues to perpetuate the predatory segmentation of the market, then our technically fair model can still exacerbate inequities. A model is not a standalone entity; it integrates into a larger ecosystem and interacts with other systems.

Trap 5: The Solutionism Bias - When ML Isn't the Answer

Not building a model at all is sometimes the most ethical decision to make. As AIML students, we naturally try to find ML solutions to every problem. However some problems are to much value and politically sensitive or have to many human factors that are unquantifiable for ML in today’s world. Building a model might give the erroneous impression of objectivity, obscuring far more serious and deeper ethical problems.

RESPONSIBLE AI

Ethics, Fairness, Transparency





Understanding the Real Risks of Superintelligent AI

Ishan Sharma,AIML 2024-28

In my class conversations always come back to AI as in Artificial Intelligence, or the sci-fi dream of a super-intelligent system that can write a human-level essay, and then multitask and fix a leaky faucet. Some of the people can't wait, and have a grin on their face like its Christmas morning. For others, it looks as if they have just heard a horrible siren and are convinced it could arrive any second.

The Present AI Reality: Flawed and Imperfect

I believe it is a distraction we should be focused on the far-off future that we miss the glitchy, imperfect AI standing in front of us. AI today is far from all-knowing, and it hasn't mastered even the simple task of reading a messy handwriting. AI has the power to change the paradigm, and most of the time, they just sit around waiting to make a big mistake. To make matters worse, they can be downright unjust and will blame you for things you haven't even done.

The Real Threat: Biased Data and Human Decisions

This is not about a Skynet-like apocalypse and also not about killer robot scenarios. We're facing college-application systems driven by biased algorithms and social media AI promoting outlandish conspiracies designed to keep users hooked and we are considering chatbots that convincingly claim that products of 7 and 8 equal 54. The super-intelligences that sci-fi plots worry about, with glowing screens, are not the real threat. It is the flawed AI that we interact with daily that pose the greater risk. It is not the machines that are the problem but rather the biased and messy data provided to the algorithms.

Reflecting Society's Flaws

Such AI systems are simply a reflection of the flaws in our society including racism, sexism, and the unending stream of misinformation crossing your phone. Big tech focuses on the fantasy of a super-AI because they want to avoid discussing the real damage their products are causing, especially the avalanche of disinformation that is filling your feed. It is a diversion, like catching a fresh whiff of coffee and forgetting why you entered the room.

Focus on Real Progress

It might be time to take a step back from the AGI hype. We should prioritize improving the AI technologies we use daily in terms of fairness, explainability, and bias. The real danger isn't robots uprising; it's us building a world on faulty, uneven, and glitchy code and calling it "progress."

Responsible AI: A Path We Cannot Ignore

Riddhi Poddar, AIML 2023-27



Artificial Intelligence (AI) has become the heartbeat of today's digital world. From the chatbots that answer our questions at midnight to the recommendation systems that somehow know what we want before we do, AI is everywhere. But here's the real question, are we using it responsibly?

The whole purpose of AI was to make human lives easier, efficient, and better. Yet, when something so powerful is created, its misuse is almost inevitable. Deep fakes that distort someone's identity, biased hiring algorithms that reject candidates based on gender or race, autonomous systems that can be weaponized AI is not just technology any more, it is a mirror reflecting society's flaws. If we feed it prejudice, it will return amplified prejudice. If we give it care and fairness, it can revolutionize the future. So, who is really responsible AI or us, the creators behind it?

The concept of Responsible AI is not some fancy research jargon. It simply means building AI that is ethical, transparent, unbiased, and accountable. It means not running after innovation so blindly that we forget the consequences. What is the point of developing the most advanced model if it discriminates against a section of society? What is the point of automation if it silently steals jobs and leaves millions unemployed without alternatives? Development without responsibility is just destruction in disguise.

As AIML students, we often get fascinated by accuracy rates and performance metrics. But is 95% accuracy enough if the 5% it fails on are always the marginalized? Numbers don't tell the whole story. Behind every dataset are human beings, with emotions and lives that could be affected by a single prediction. The excitement of technology should never blind us from the humanity attached to it.

Let's think about accountability. If a self-driving car makes a mistake, who should be blamed the machine, the programmer, or the company deploying it? These are not science fiction questions any more; they are legal and ethical battles already happening. The uncomfortable truth is, AI cannot be held guilty in a court of law. The responsibility will always fall back on us. That is why transparency is so crucial.

Every AI decision should be explainable, not a black box mystery but a process that can be understood, questioned, and corrected. The role of Responsible AI is not just in avoiding harm but also in maximizing good. Imagine AI systems that predict climate disasters early and save thousands of lives. Imagine healthcare AI that helps doctors diagnose rare diseases faster, not replacing them but empowering them. Imagine education powered by AI tutors that adapt to every student's unique pace. This is the positive power AI holds if only we direct it responsibly. But responsibility is not the job of developers alone. Governments need strong regulations, companies need to prioritize ethics over profit, and as individuals, we need to question what we consume and create. Responsible AI is not one person's duty; it is a collective responsibility of society.

In the end, the question we need to keep asking is simple: Are we building technology that makes us more human or less? Because if innovation strips away empathy, fairness, and trust, then it is not progress at all. AI should not just be smart it should be wise. The future of AI is not just about algorithms; it is about choices. And Responsible AI is the choice that decides whether technology becomes a weapon of division or a tool of empowerment. The power lies in our hands.

These figures contrast their P&L numbers. Last year, they expected 5 billion dollars





The Quiet Conscience of Code: A Reflection on Responsible AI

Drishtti Narwal, AIML 2022-26

There was a time when intelligence was something we saw in people — in their eyes, their voice, their questions. But now, it hums quietly behind screens, stitched into algorithms and stored in clouds. Today, machines make decisions we once reserved for careful human judgment — who qualifies for a job, which patient needs urgent care, which post gets seen. It's impressive. It's efficient. But in the middle of all this brilliance, a quieter thought echoes — are these machines learning wisdom, or just mimicking patterns?

Responsible AI is not just another buzzword on a project charter. To me, it feels more like a gentle responsibility — a moral thread we must not lose hold of, no matter how fast the tech races ahead. Behind every model, every decision, is a trail of human choices: the datasets we trust, the goals we define, the edges we ignore. Algorithms don't just reflect reality — they shape it. And sometimes, they shape it wrong.

Imagine an AI trained to predict crime — but the data it's fed is riddled with historical prejudice. It doesn't learn safety. It learns suspicion. Or a hiring tool that screens resumes and filters out brilliance, just because it didn't recognize it in the 'right' format. The problem isn't always in the code — it's in the silence around the code. In the blind spots we leave unchallenged.

Responsible AI means not looking away. It means asking hard questions even when the numbers look good. It asks for engineers who care, researchers who question, leaders who listen, and communities who are heard. It's about building things that don't just work, but work right.

I've come to believe that we aren't just programming machines — we're projecting our values onto the future. Every dataset, every design choice, is a piece of our collective conscience. We can't outsource ethics to automation.

So, we must pause and ask: *What kind of world are we building? What kind of justice, what kind of empathy, what kind of inclusion does our technology reflect?* Because code has no heart. We do. Responsible AI isn't about resisting change. It's about steering it — with humility, with intention, and with the people who will be most affected, sitting at the table. It's about ensuring that the future we automate doesn't come at the cost of the very humanity it was meant to serve.

AI with Integrity: Operationalizing Responsibility for Real-World Impact



Dr Gargi Bhide
Assistant Professor,
AIML Department

Responsible AI is NOT the same thing as Responsible Data Use or Ethical AI. Responsible Data Use is about how you go about collecting, storing, utilizing and sharing data that prioritizes individual privacy, data security, and fairness (i.e. non-bias). Ethical AI is about the development and deployment of AI systems that prioritizes fairness, transparency, accountability, and respect for human values.

Responsible AI seems to marry both Responsible Data Use and Ethical AI, while adding on elements of system robustness.

Core Principles of Responsible AI-

1.Explainability — accurate prediction, traceability, how the AI makes decisions.

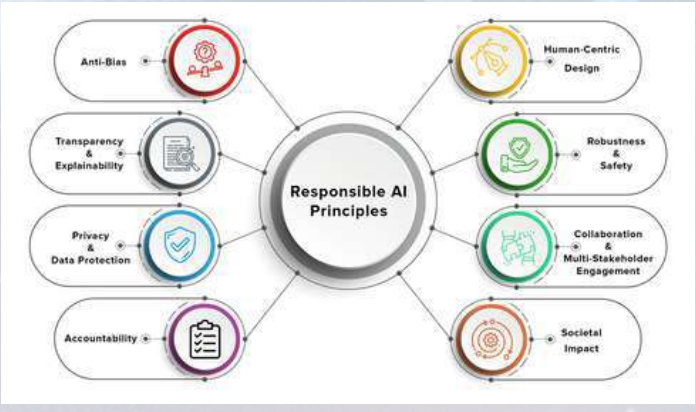
2.Fairness — diverse & representative data, bias-aware, bias-mitigation, developed by diverse talent, subject to ethical review.

3.Robustness — not brittle when encountering (intentional & unintentional) anomalous data, strong cyber and digital security.

4.Transparency — end-users must be able to see how it works, evaluate the strengths and weaknesses of its features and functionality.

5.Privacy — protection from leakage or misuse of personal data.

Responsible AI Operational Framework = Model Governance + Experience Design + Diagnostic + Integration.



AI IN GENERAL

The Future of Intelligence



Case Study: How Airbnb Scaled the AI Wall to Revolutionize Travel Search

Kunal Jindal, AIML 2024-28

The Business Problem: Hitting the Plateau

Every tech company values growth above all else and during the early years of the company, Airbnb was no different and relied mostly on a type of machine learning known as GBDT or Gradient Boosted Decision Trees. This model became a machine learning workhorse for Airbnb, consistently and successfully matching millions of guests to the accommodation of their choice and propelling the rapid growth of the company. Every engineering team, however, comes to a wall and in this case, it was a plateau in performance. There was a clear and simple business challenge in this situation, which was to find a way to surpass a mature ML system and to provide the next generation of personalization and relevance in search.

The Strategic Solution: Betting on Deep Learning

For Airbnb, the solution involved a lengthy, large-scale investment in the appropriate advanced deep learning technologies. This effort meant more than switching models; entire components of their AI architecture and overarching philosophy had to take a 180 degree turn. The aim at the time was to construct a neural network sophisticated enough to pinpoint the intricate aspects and nuances of customer behavior something the previous models failed to do.

Every phase of the Deep Learning pivot was painstakingly strategic. Beginning with a functional, "basic" neural network served as the proof-of-concept to assess whether their production systems could support the new frameworks.

The first marked advancement was the introduction of a "Lambdarank Neural Network." What transformed the training focus of this model was the ability to work within a specified construct. Beyond predicting whether an individual listing would be booked, it was designed to assess and score pairings. It assessed, "A guest books Listing A and skips Listing B. What changes in ranking should occur?"

The model worked to within business optimization frameworks, solving the challenge of positioning the most relevant results at the top of the listing. Most importantly, it positioned the listings to within the thresholds of actual bookings driven by the listing.

Overcoming Deep Learning's Quirks

Adopting deep learning on such a massive scale came with its own set of unique challenges. The team discovered that simply feeding raw data into powerful neural networks wasn't enough; the data needed careful preparation. Features like listing price, historical booking count, and geographical location had to be "normalized" and presented in a "smooth" distribution for the neural network to learn effectively.

"One of our biggest lessons was that deep learning isn't a magic bullet you just drop in," Dr. Singh noted. "It's an entire ecosystem shift. We had to rethink how we engineer features, how we train our models at scale, and even how we understand what the AI is 'thinking'. This process pushed us to investigate higher-level problems, rather than getting bogged down in manual feature tuning."

For instance, initial attempts to use unique "listing IDs" as direct inputs led to "overfitting," where the model became too specialized in the training data and performed poorly on new searches. The sheer volume and sparsity of unique listings meant the model couldn't generalize. This led to innovative solutions, including re-engineering how geographical data was fed into the system, allowing the AI to learn preferences for entire neighborhoods rather than specific coordinates.

To handle the immense data, Airbnb engineers dramatically sped up their training pipeline by a factor of 17, enabling the system to process billions of search interactions, moving from weeks of training to just months.

A Future Shaped by AI

From a student's viewpoint, the Airbnb case study showcases applied machine learning. It captures the essence of success in industrial AI: It's not about grabbing the latest algorithm.

- Understanding technological plateaus, then courageously betting on a new paradigm.
- Tackling the engineering puzzles involving data prep, model generalization, and the scaling of computation.
- Ensuring alignment of the technical models to business metrics, as in the case of the Lambdarank approach where the optimization focus was top-of-list relevance.

Airbnb's story demonstrates the power of deep learning and engineering focus to shift the business axis to the user. This approach unlocks value and transforms the core experience of the product for millions.

Case Study: How Google Solved its Machine Learning Factory Problem with TFX

Sreevals Nair, AIML 2024-28

The Industrial Challenge: The "ML Prototype to Production" Gap

In academic projects, we often focus on building a single, successful model. However, in a large company like Google, the real challenge isn't building one model—it's building, deploying, and maintaining hundreds of models reliably at a massive scale. This is known as the "production ML" problem.

Google faced a critical bottleneck: moving a machine learning model from a researcher's laptop to a live product serving billions of users was a slow, error-prone process. Different teams used ad-hoc scripts, leading to:

- **Technical Debt:** Fragile, one-off pipelines that broke easily.
- **Inconsistency:** "Training-serving skew," where a model performed well in testing but failed in production because the live data was processed differently.
- **Slow Innovation**: It could take months to test a new idea and deploy it, stifling innovation and rapid improvement.

The business need was clear: create a standardized, scalable platform that could turn experimental ML into a reliable, industrial-scale process.

The Strategic Solution: TensorFlow Extended (TFX)

Google created TensorFlow Extended (TFX), a fully integrated machine learning (ML) platform designed to function like an assembly line for machine learning. Rather than having researchers and engineers construct custom pipelines for every project, TFX offers a collection of standardized, reusable parts. The primary understanding was that the success of an ML system involves much more than the model; it revolves the surrounding ecosystem. TFX was designed to handle this surrounding ecosystem, which it divides into a number of primary phases.

1. The Data Foundation: Fixing the "Garbage In, Garbage Out" Problem What might be the most surprising lesson from this case study is that a lot of production ML is not about the models but rather about the management of the data. TFX has very powerful components that focus entirely on the health of the data.

Data Validation: This is a sort of continuous quality verification system for data. It automatically checks incoming data against a "schema" (a pre-defined set of rules) and identifies anomalies. In the case of a "user age" feature, if the number starts to show negative values, the system identifies this error quickly before it adversely impacts the model.

The Google Play Breakthrough: In the Google Play store case study, the data validation component uncovered a catastrophic error: certain features that were supposed to be present during model training were persistently absent in the actual environment. Closing this data gap was enough to realize a 2% improvement in app installs. Such a significant business impact was the result of simply improving data quality.

2. Industrial-Grade Model Training

TFX has started to implement intelligent features that focus on optimizing training with efficiency and dependability at scaling.

Warm-Starting: TFX saves a great deal of training time and costs. Instead of training an entire new model from scratch each time, TFX "warm-starts" by partly reusing elements from a previous model (like embeddings). This enhances the possibility of frequent updates of the models to accommodate new data.

High-Level APIs: TFX has incorporated industry standards and best practices into its system, simplifying the construction of production-ready models even for engineers with little expertise in the low-level particulars.

3. Rigorous Model Evaluation and Safeguard

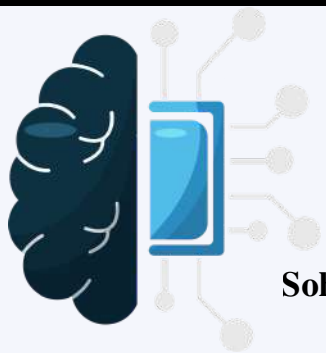
Sliced Evaluation: The platform doesn't just look at overall model accuracy. It automatically evaluates performance on specific slices of data (e.g., "users in Japan" or "users on Android devices"). This helps catch performance regressions that would be hidden in the overall average. Automated Validation: Models are compared against a baseline and must pass predefined quality gates before they are allowed to be deployed. This prevents a bad model from ever reaching users.

4. High-Performance Serving with TensorFlow Serving

Multitenancy: A single server can host multiple models at once. To prevent a slow model update from slowing down all other live models, Google implemented dedicated resources for loading new models, reducing latency spikes by over 10x.

Optimized Data Parsing: They developed highly efficient code to read the incoming data, achieving a 2-5x speedup in data processing—a critical improvement when handling billions of requests.





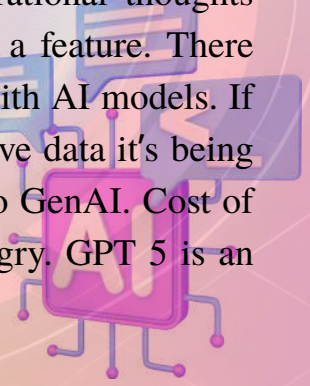
OpenAI: A Case Study in the Economics of GenAI

Soham Kulkarni, AIML 2023-27

We've had a couple of AI revolutions till now, each having geniuses at the helm, claiming that AI can replace humans. A machine which can think. A machine which can think better than a human, maybe a machine smarter than you. This vision is scary, playing host to several brilliant works of science fiction. History tells us that AI doesn't grow gradually. Because this fear can only propel the technology so far. A market correction has followed every AI revolution, where people realised that the technology, while brilliant, had limited scope for use. An AI winter, if you're being dramatic. Is this time any different? GenAI proponents claim that the state-of-the-art models can generalise.

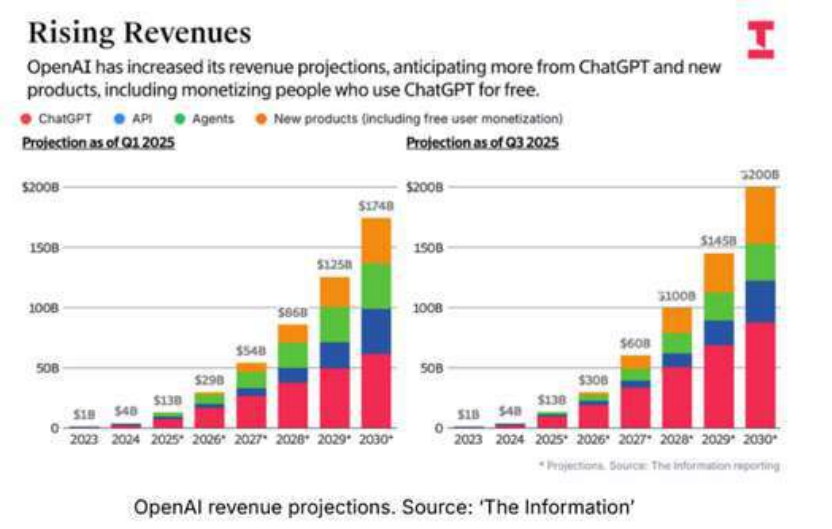
Proponents claim that these technologies have limitless scope of use instead. We've been seeing an incredible amount of hype surrounding GenAI. This hype manifests itself in several ways. Commonly observable effects include very high usage figures reported by AI platforms, job freezes and mass layoffs, and most importantly, mindblowing amounts of investment.

OpenAI can be considered to be one of the industry stalwarts for this newly born domain. I'll be focusing on them, as they're a purely GenAI company, unlike other players which have significantly diversified revenue streams such as advertising Google and Microsoft). The digital economy allowed for a completely new business model driven by data. Large companies can afford to subsidise, or offer their technological solutions for free. In return, they get access to massive amounts of data, which they can sell, use for training advertising algorithms, and to build shadow profiles. Even if you′ve never signed up for Facebook, the company still has a general sense of who you are, gathered through uploaded contact lists, photos, or other sources. It's not as easy with GenAI data though. OpenAI has had issues with sycophancy in GPT 4, because live data indicated users wanted more agreeable models, even if the generated content was incorrect. This was an especially dangerous ‘bug′ affecting users who rely on ChatGPT for emotional assistance. ChatGPT can serve as an echo chamber for one′s irrational thoughts because user data indicated that ChatGPT being agreeable over objective is a feature. There have been cases of suicide where the victims had codependent relationships with AI models. If a GenAI solution can cause such large scale societal damage, just based on live data it′s being improved with, the data-driven subsidisation business model may not apply to GenAI. Cost of inference per token may be dropping, but models are increasingly token hungry. GPT 5 is an excellent example of the cutting edge token hungry approach.



With their StarGate datacenter project, OpenAI has managed to overleverage one of the largest investment groups in the world, the SoftBank Group. This 40 billion dollar deal is the world′s largest private tech fundraise, valuing OpenAI at a whopping 300 billion dollars. Critics have said SoftBank hasn′t put so much money on one horse even with their investment in Arm. If we analyse their revenue projections, it paints a quite promising picture. This picture aligns with the limitless scope of use narrative the current AI revolution promises.

OpenAI revenue projections. Source: ‘The Information′



OpenAI revenue projections. Source: 'The Information'

These figures contrast their P&L numbers. Last year, they expected 5 billion dollars of loss on 3.7 billion dollars of revenue. Their projections also starkly contrast their user conversion ratio. OpenAI has over 700 million active users per week as of August 2025. They crossed 2 million business users in February 2025. That′s a low conversion rate. The cost of training is increasingly exorbitant, and model performance is not scaling with training cost. Can pure GenAI model training/hosting be a profitable business model? I'd argue no. Models training doesn′t have a moat, and hosting costs have to be passed on to users somehow.

Sam Altman has publicly admitted they lose money even on their 200$/month Pro tier . Recently, there was a press release from OpenAI launching ‘ChatGPT Go′, their cheapest yet plan, launched exclusively in India at 400₹/month . Sam Altman has stated that India is their second largest market after the USA , growing fast. Most conventional businesses are often worried about entering the Indian market. We're some of the most finicky consumers, with very little customer loyalty. We expect the world from any product, and we′re only willing to pay spare change for it. Large names come and go, unable to weather our cutthroat economy. A third world country with first world consumers. Monetising their user base is going to be an incredibly difficult task for OpenAI, even though ChatGPT started out as the fastest growing application in the world.

Application of Artificial Intelligence in the Upcoming Areas of Two-wheeled Electric Vehicles Technology

Vishwas Vaidya , AIML 2024-28



1.0 Introduction:

As automotive world is being penetrated with electrified power-train, the same is getting augmented with connected and autonomous driving techniques. The common bond that binds this trinity is nothing but artificial intelligence (AI) . In a way this strategy mimics a human being travelling on foot. The human intelligence makes this mode of transport autonomous. Human senses such as vision, hearing, skin etc. further powers us to negotiate dangers, communicate with other road-users etc. in accomplishing safe and efficient transport. This article highlights role of AI.

2.0 AI Powered Self Balancing Bike/Scooter:

For a developing country like India, it makes sense to work towards "Autonomous and connected two - wheeled vehicles". It necessarily has to be "self-balancing". The AI will emulate inverse pendulum logic to lend self-balancing support to the vehicle. The vehicle will communicate with other road users and connected infra-structures such as street lights, traffic lights, pedestrians, zebra crossings. Each of the infra-structure element will be equipped with AI. This "distributed AI" will achieve what even four-wheeler-based AI has not been able to achieve so far. It can predict accidents, minimize traffic jams, and self-navigate the riders to their destination.

3.0 Pre-empting Battery Burning Incidences:

Major reasons behind battery burning is: electrochemical operational abuse of the battery during vehicle traction. In other words: if the manually driven EV is aggressively accelerated and decelerated, the battery may age faster and get heated progressively. An autonomously driven vehicle will precisely address this point by ensuring calm and quiet style of driving with near uniform vehicle speeds for most of the scenarios. This will also enhance the battery life. AI can progressively track the battery state of health and deliver a predictive alert for the rider regarding impending fire a few minutes before in extreme scenarios. Hence the vehicle users can easily abandon the vehicles to prevent such mishaps.

4.0 Concluding Remarks:

Autonomous and connected two-wheeled EVs powered by AI is a neglected area of research worldwide comparatively. This article tries to highlight relatively untouched opportunities powered by AI in this area. Another area where AI can contribute is smart charging of the vehicle by adopting the charging methodology to the "personality" of individual vehicle systems. We in developed world need to focus our energies towards developing these areas strongly.

Your AI Advantage: Why Mastering Collaboration is the Ultimate Career Move

Shubhankar Bhan, AIML 2024-28



As an undergraduate studying Artificial Intelligence and Machine Learning, I'm constantly surrounded
by conversations about the future of work. The most common fear I hear, from classmates and family alike, is that AI will take our jobs. But the more I learn, the more convinced I become that this fear is misplaced. AI isn't the threat; complacency is. In my view, it's not AI that will replace you—it's someone who knows how to use AI better than you do.

AI isn't a shortcut—it's a lever

Let's be honest, technology has always changed the way we work. From the steam engine to the internet, every major leap has caused anxiety. But history shows that those who adapt and learn to use new tools not only survive—they thrive. The same is true for AI. The people who are most at risk are not those whose jobs are "automatable," but those who refuse to learn and evolve.

In my classes and projects, I've seen firsthand how AI can turn a daunting task into something manageable, even enjoyable. Whether it's automating tedious data collection or generating creative ideas for a project, AI is like having a supercharged assistant at your side. I've seen classmates who struggled with coding suddenly build impressive projects with the help of AI-powered code assistants. Creative friends use AI to brainstorm designs or write stories. Far from making us obsolete, AI makes us better at what we do. It's like giving everyone a superpower—if they choose to use it. It multiplies what you can do. The students who embrace it are already pulling ahead, not because they're smarter, but because they're working smarter.

The Human Touch Still Matters

Some worry that AI will make human skills obsolete or will make everything robotic and impersonal. I disagree. The best results I've seen come from people who blend AI's strengths with their own creativity and empathy. For example, in a recent group project, we used AI to analyses large datasets, but it was our team's unique perspective that turned those insights into a compelling story. AI did the heavy lifting, but it was the human touch that made the work meaningful.

In fact, I think AI makes human qualities even more valuable. Creativity, empathy, critical thinking—these are things AI can't replicate. But AI can free us from repetitive tasks, giving us more time to focus on what truly matters. For example, in healthcare, AI can analyse scans and flag anomalies, but it's still the doctor who talks to the patient, makes the final call, and provides comfort. In journalism, AI can summarize data or suggest headlines, but it's the human writer who crafts a compelling story. The magic happens when humans and AI work together, each doing what they do best.

Learning to Learn: The New Competitive Edge

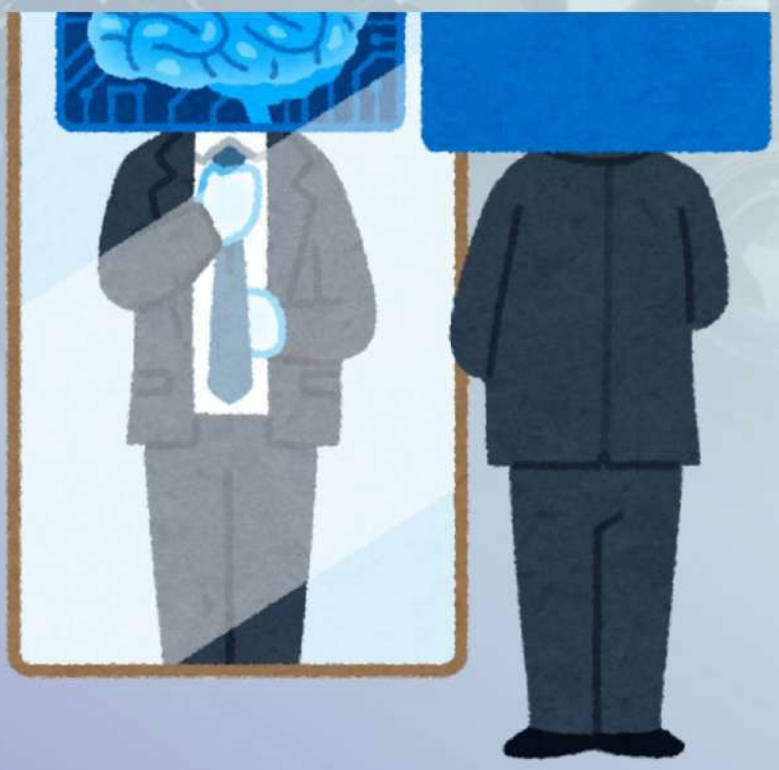
If there's one thing I'm learning as an AIML student, it's that the most important skill isn't knowing a specific programming language or algorithm—it's learning how to learn. AI is evolving so quickly that what's cutting-edge today might be outdated tomorrow. The people who will succeed are those ho stay curious, keep experimenting, and aren't afraid to ask questions.

I think about my own journey. When I first started, I was intimidated by all the new tools and concepts. But once I started playing around with them—building small projects, testing out APIs, even just reading AI-generated summaries of research papers—I realized it's not about being perfect. It's about being willing to try, fail, and try again.

A Mindset Shift: The Choice Is Yours

Here's my honest opinion: AI is not a threat to your job unless you let it be. If you ignore it, refuse to learn, and cling to the old ways, then yes—someone who embraces AI will outpace you. But if you're willing to adapt, to see AI as a partner rather than a rival, you'll find opportunities you never imagined. I'm excited about the future, not afraid of it. I see AI as a tool that can help me do more, think bigger, and solve problems I couldn't tackle alone. The real danger isn't AI—it's refusing to grow alongside it; it's missing out on the opportunities it creates.

So, I urge all my fellow future professionals: Don't fear AI. Learn to use it. Experiment with it. Make it your ally. That, in my view, is the surest path to success in the age of artificial intelligence. And that person could be you.



From DDLJ to Deep Learning: Generations Change, Algorithms Stay



Dr Pooja Kamat
Associate Professor,
AIML Department

1995. Dilwale Dulhania Le Jayenge hits the big screen. A generation learns about love, train chases, and the magic of "palat… palat."
2025. Students are learning about transformers, neural nets, and deep learning.
Different eras. Different obsessions. But one thing hasn't changed: algorithms still run the show.

The DDLJ Generation — and the 2005 Engineer in Me

When I started my Computer Science and IT engineering journey back in 2005, the world was simpler. C++ was our comfort zone. "Cloud" meant the thing outside your window. We debugged line by line, carried USB drives everywhere, and prayed our code would finally compile without errors.
There were no AI assistants to explain code or fix logic — just you, your computer, and sheer determination.

The Deep Learning Generation

Today's students are in a completely different universe. They learn concepts through YouTube shorts, use AI tools to test logic, and sometimes have more computing power on their phones than we had in our labs.
Learning is faster, smoother — and sometimes, dangerously, too easy.
But Under the Hood…
No matter how much the tech evolves, the fundamentals haven't changed.

- Sorting is still sorting — whether it's on paper or in Python.
- Regression is still regression — just wrapped in fancier names.
- Optimization still means trade-offs — whether in Dijkstra's algorithm or your Ola route.

Generations change. Algorithms stay.

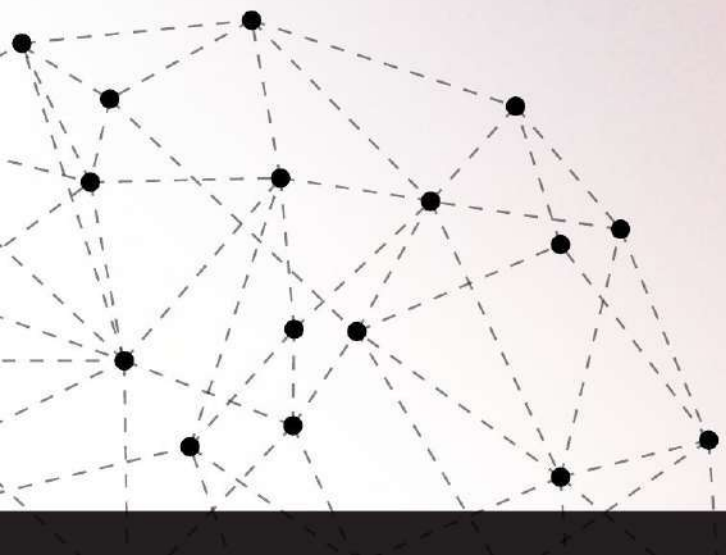
If Bollywood Had Algorithms…

Sometimes I think: if Bollywood were code —

- Overfitting would be like adding too much drama — looks great, loses purpose.
- Regularization is the editor trimming unnecessary scenes.
- Gradient Descent? That's Raj running after Simran's train — chasing the "minimum" distance before it's too late.

The Bigger Picture

Every generation thinks their tools make them smarter. But whether you're debugging in Turbo C++ or training a transformer model, what really matters is the same: logic, patience, and curiosity. Technology evolves. The engineer's mindset — the desire to break, build, and understand — is timeless. And maybe that's the real DDLJ moment of learning: You don't chase the train because it's fast. You chase it because you understand why it's worth catching. What's your "DDLJ to Deep Learning" moment? Something old-school you learned that still shapes how you think in this AI era?





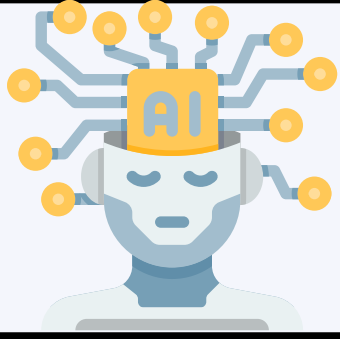
GENERATIVE AI

Crafting New Realities



Deception in AI: How Trickery Drives Creative Breakthroughs

Akshita Mishra,AIML 2024-28



You might find AI generated art, deepfake videos, or AI composed music all over the web. But you might wonder , how these digital artists and devisers learn to create such realistic forgeries. I recently went over a complex and advanced paper from 2014 by Ian Goodfellow and Yoshua Bengio that introduced the concept of Generative Adversarial Nets , or GANs for short. This concept serves as the foundation for a lot of the extraordinary AI technology today this paper serves as an origin story for a lot of the AI technology we are currently using in our daily lives. I was like the paper was trying to explain quantum physics to a dog. But the main idea of the paper was a huge game of cat and mouse, or in this case art forgers and art detectives, where forgers are trying to evade the super smart art detectives trying to figure out how to expose the forgery and catch the forger.

The Ultimate AI Showdown: Forger vs. Detective

Why all the fuss about GANs?
Well, picture these two AI systems locked in an endless duel, with one trying to outsmart the other. It is precisely this competition that gives them their strength.
First, the generator (G) :- Imagine it as a master art forger. Forger produces images, text, and music that are so close to real that it is indistinguishable. Initially, G creates worthless images, just blobs, but it learns.
discriminator (D) :- It is the art detective trying to identify the forger's fakes G produces and the real ones (the training data). It detects G's fakes and detects G. It's a never-ending hunt. The genius of GANs is that both systems train simultaneously. G tries to fool D, while D gets better at spotting slips. It's a never-ending loop that drives both systems to improve.

How This Digital Dance Works

The Generator seeks to fool the Discriminator into error to classify bogus data as authentic. In contrast, the Discriminator focuses on the data that counterfeiter generates and seeks to correct the classification. The original paper refers to this back-and-forth as a "minimax two- player game." (It's a video-game-like scenario, or a rather game of chess of two AIs) Both parts are constructed of multilayer perceptrons, which is a convoluted term for deep neural networks.
The most interesting bit is that they are trained using backpropagation, which is the most common way to teach neural networks.

Why should we care is because older methods of data generation were truly cumbersome. Things like Markov chains or approximate inference networks were used basically the slowest and most complicated tools available for training. GANs don't need any of that. Simply provide the data and let the Generator and Discriminator compete against each other until the model improves.

What is the target for the Generator to fool the Discriminator into thinking its output is real, at which point the Discriminator is just guessing and is unable to tell the difference even 50 percent of the time. The Generator has captured the data distribution and can now produce a virtually limitless supply of realistic counterfeits. Imagine an endless supply of cat photos, unseen fashion, and architectural structures that exist only in imagination.

Why GANs Are a Game Changer

- No More Markov Chains (Bless!): Older generative models relied on complex and slow methods like Markov chains, which often produced blurry, low-quality outputs. GANs skip all that nonsense and still get better results.
- Simple Training: No need for wild hacks or complex math. Just good ol' backpropagation — the standard way we train most neural networks. Easy to implement, easier to scale.
- Super Flexible: You can swap in different types of networks for both the Generator and Discriminator. CNNs, transformers, you name it. GANs can be tailored to tons of different tasks.
- Crisp, Realistic Outputs: Because the Generator is trying to fool the Discriminator, it gets better at generating high-quality, sharp images — way more detailed than what earlier models could do.
- Learns Patterns, Not Just Memorization: The Generator never sees real data directly. It only gets indirect feedback from the Discriminator. That forces it to learn the underlying structure of the data instead of just copying examples.

Training GANs is more of an art than a science. It's a balancing act:

- If the Generator gets too good, too fast, it might find one trick that works and just stick to it — like only producing one style of image (this is nicknamed the "Helvetica scenario", like using just one font over and over).
- If the Discriminator gets too powerful, it can crush the Generator before it learns anything useful. Training collapses, and you're back to square one.

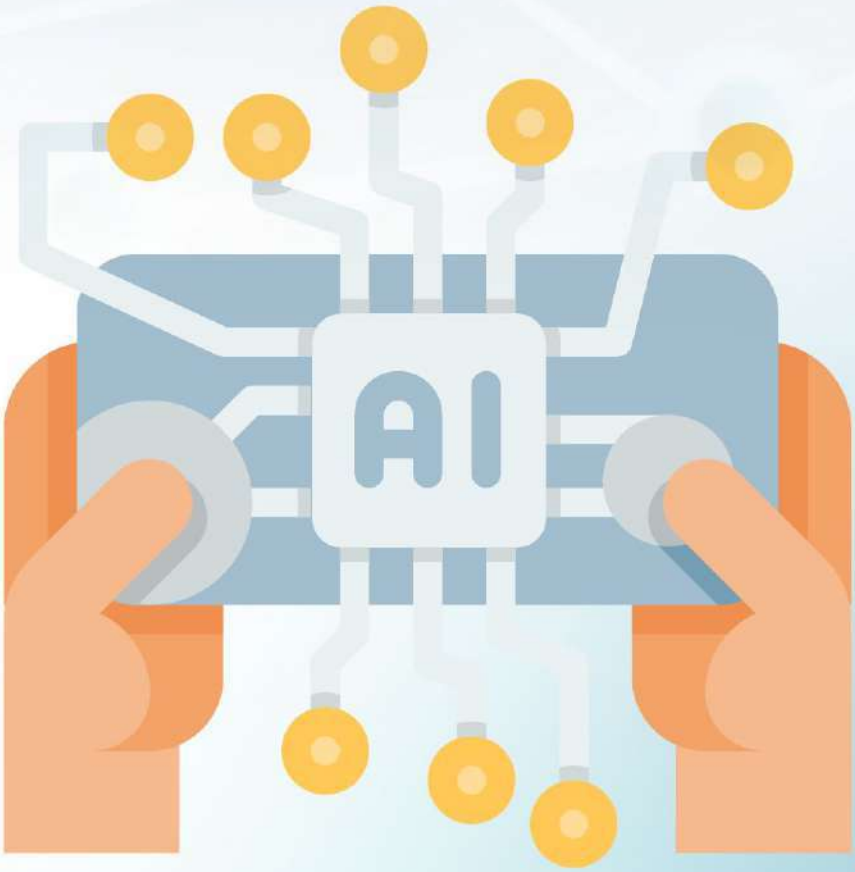
And here's the wild part the images they showed weren't cherry picked they were random samples straight from the Generator. No curating, no hand picking the best ones to make the model look good. For a brand new approach at the time, that was a huge deal. It hinted at just how powerful this idea could become once researchers started improving and scaling it. And, well… we've all seen how far GANs have come since then.

What's Next for These Digital Deceivers?

The researchers behind GANs knew they had something special. They wrapped up their paper by pointing to some exciting possibilities that, at the time sounded futuristic but are now totally real:

- Conditional GANs: Imagine being able to ask the AI for something specific, like "generate a picture of a red car" instead of just a random car. (this became a huge deal in models like StyleGAN and DALLE)
- Smarter Training: They envisioned better ways for the Generator and Discriminator to train together making them not just rivals, but effective co-learners.
- Semi-Supervised Learning: Even with just a little bit of labeled data, GANs could help other models learn more efficiently. Huge boost for situations where data is limited.

So next time you see a hyper-realistic fake photo, deepfake video, or a jaw-dropping AI artwork, remember it all started with a clever idea treat AI training like a game between a forger and a detective.



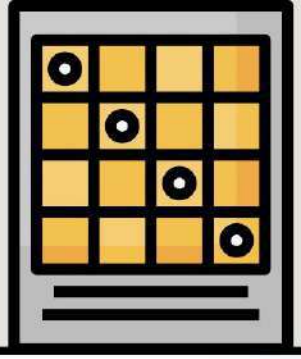
NEURAL NETWORK

The Architecture of Intelligence



Understanding the Lottery Ticket Hypothesis in Neural Networks

Kashif Kamran, AIML 2024-28



I just finished reading Frankle and Carbin's paper The Lottery Ticket Hypothesis for my machine learning class, and now I'm questioning much of what I thought I knew about neural networks like pulling a thread and watching the whole sweater unravel.

At first it sounds simple, but once you see it, it turns standard practice on its head like swapping the usual black coffee for a burst of bright, lemony tea. We've all grown up hearing that bigger wins, like choosing the largest slice of cake just because it's there. Want sharper results? Build it up with more layers like stacking warm blankets on a cold night. Is your model hitting a wall? Turn the knob and bump up the parameters.

But that creates a real hurdle for students like me, stuck without the roar and hum of a massive server farm. These huge models crawl along, feel clunky, and rely on brute force like trying to crack a walnut with a sledgehammer. We also learned about pruning, where you trim pieces off a large, trained network like snipping extra branches from a tree to make it smaller. It always felt like a quick patch, like sweeping crumbs under the rug after everyone's gone.

Training a tiny, stripped-down network from scratch felt hopeless; every time I'd given it a shot, it crashed and burned, like watching sparks fizzle out in the rain. That's why the Lottery Ticket Hypothesis struck me it landed like a sharp knock on the desk.

It's as if every sprawling, randomly built network hides a tiny, flawless "winning ticket" inside like finding a perfectly carved key in a dusty attic. It's not about cutting away the excess once the work's done it's about uncovering the golden blueprint that's been hiding in plain sight.
What really blows my mind is the "reset" step in their method, like wiping a chalkboard clean in one swift motion. You don't just spot the connections and hang on to whatever numbers they've been trained with. You set them back to their original random values, like tossing a handful of dice and starting fresh. It's the exact reverse of everything we're taught to believe like being told the sky is green. They say the learned weights are what really matter like the numbers quietly steering a model's decisions. This shows that for these subnetworks, the spark comes from starting in just the right spot and pairing it with the perfect structure like striking a match on coarse stone. It's more than the path it's the seed in your hand and the map tucked in your pocket.

What this means for a student like me is huge:

1. Accessibility: Discovering small but powerful networks that can train in the blink of an eye could
change everything. It could open up cutting-edge research to people who don't have huge computing budgets, from small university labs to independent learners hunched over laptops in quiet cafés.

2. Smarter Design: It makes us stop and wonder are we putting our networks together all wrong, like stacking bricks that don't quite fit? Rather than simply building ever-larger models, maybe we should figure out how to pinpoint these efficient subnetworks right from the start like spotting a single bright thread in a tangled skein. It hints at a future where architecture is sleeker, smarter, and as precise as a pencil line drawn on fresh paper.

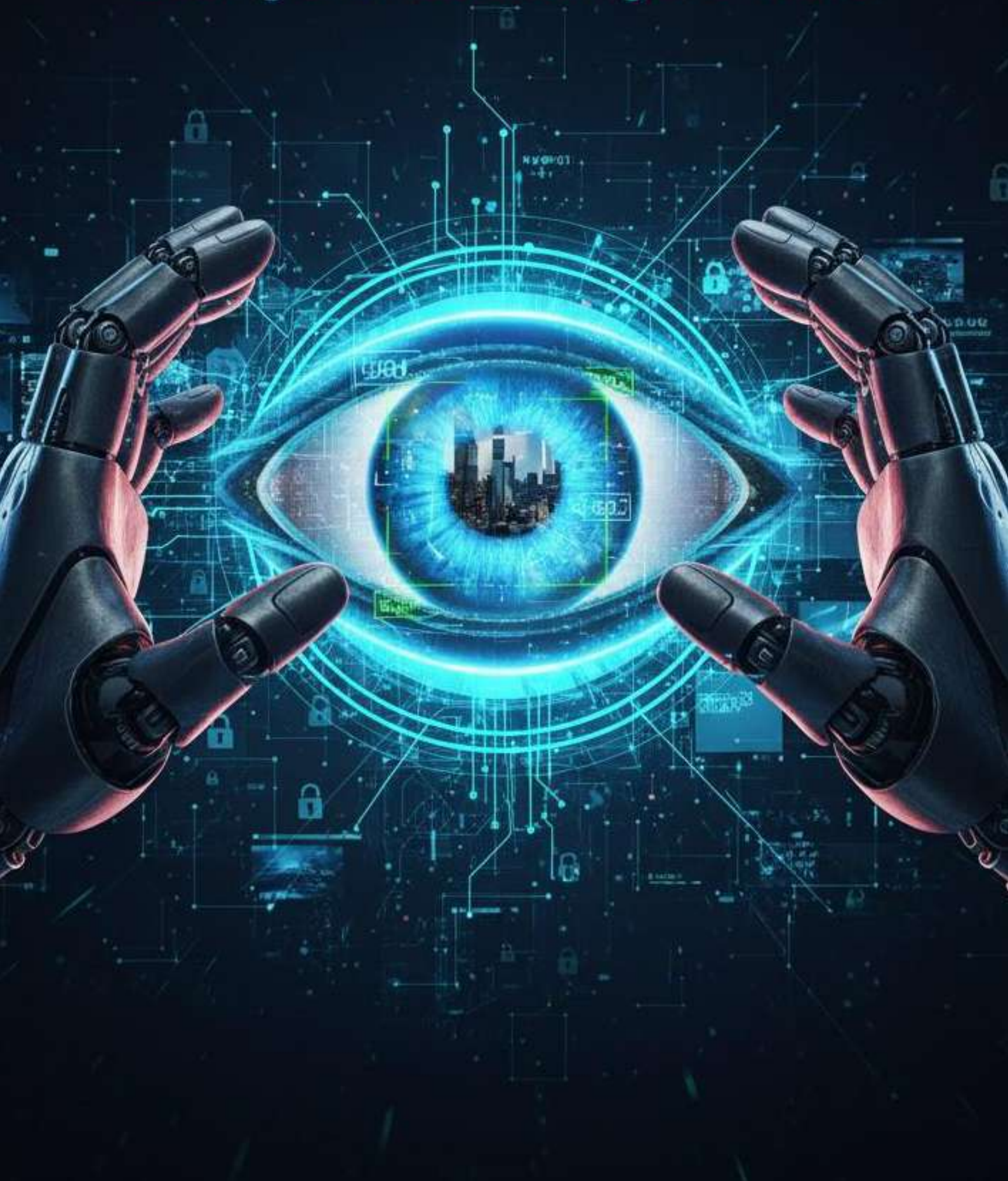
3. A Deeper Mystery: Honestly, this hypothesis is thrilling and it's got a way of making you feel small, like staring up at a night sky full of stars. It's a reminder that we still don't truly grasp why deep learning works so well or exactly how it pulls off results that can feel almost uncanny. These "lucky" starting setups hint that big models may owe their success less to raw power and more to the vast search space they offer for finding those perfect, razor-thin subnetworks.

In my perspective, the Lottery Ticket Hypothesis is far more than a parlor trick for constructing smaller models. It is a very important challenge to our fixation on size. It heralds the possibility that
powerful, more efficient, and sustainable artificial intelligence may not necessitate the construction of larger models but will derive from learning how to extract the more efficient and hidden "gems" caught inside. This feels like the more intelligent and more interesting way to proceed for the future
of artificial intelligence.



AI IN COMPUTER VISION

Seeing & Understading the World



How NeRF Is Changing the 3D Rendering: A Look at the Future of Graphics

Abhishek Shrivastava, AIML 2024-28

It's one of the most exciting things I've had the opportunity to work with in a long time. Every year I see advances in the technology that offers the possibility of generating a set of photorealistic 3D scenes from real world images, but it still seems to be an unattainable goal. I have generated complex 3D meshes, tried using point clouds, but still seem to be in a tedious and overwhelming process of generating realistic scenes. Not long ago, I came across a paper published by researchers from UC Berkeley, Google Research, and UC San Diego. The paper includes something called NeRF: Neural Radiance Fields for View Synthesis, and I must admit, it is a huge leap. NeRF is a other way to encapsulate, and for lack of a better word, mimic a 3D scene, and it is certainly groundbreaking. It defies a lot encapsulate, and for lack of a better word, mimic a 3D scene, and it is certainly groundbreaking. It defies a lot of paradigms of the 3D scene of photorealistic rendering. It may be the step we need to streamline the most photorealistic rendering.

What is This NeRF Wizardry?

NeRF fundamentally departs from traditional 3D modeling methods which involve working with polygons or voxel grids. Instead, it leverages a fully connected neural network, or a multilayer perceptron, to capture the entire scene. It certainly is an idea worth exploring. NeRF learns what is called a continuous 5D radiance field. What does it mean is for every point in the 3D space (x, y, z) and every viewpoint (θ, φ) the neural network produces two results.

1. What color light is emitted from that point when viewed from that direction (RGB).

2. How dense or opaque that point is (volume density).

NeRF applies traditional volumetric rendering methods, similar to illuminating a foggy room with a flashlight and watching how the light is blocked and scattered. The conclusion is that NeRF can create completely new, photorealistic views even from angles that were not a part of the original photos. It is almost like a virtual camera can be set to any position in the scene and a highly realistic image would be produced.

The fact that the complete process is differentiable makes it even more impressive. In layman's terms, that entails the model is capable of self-correction. Given a few photos of a scene and respective camera positions NeRF is able to modify its internal representation iteratively until the images it produces align as closely as possible to the actual photographs. In comparison to older techniques, it maximally enhances the optimization in pursuit of visual precision.

How NeRF Achieves Realism:

1. Positional Encoding: Capturing Fine Details

One major drawback of basic MLPs is that they learn smooth, low-frequency functions. This is a problem because real-world scenes have sharp discontinuities, fine textures, and reflective surfaces. Researchers have proposed the use of positional encoding to alleviate this problem. In this case, 3D coordinates and viewing directions are transformed using high frequency sine and cosine functions prior to being fed into the network. This allows the MLP to capture far more detailed and intricate variations present in the scene. Without this, the output is overly smooth and blurry, as is the case in the comparison shown in Figure 4 of the paper.

2. Hierarchical Volume Sampling: Smarter Rendering

Volume techniques often involve sampling points uniformly along every camera ray but such an approach is contextually inefficient as many samples will traverse empty space or regions occluded by other geometry. NeRF addresses this issue using a two-stage sampling method. In the first stage, the model makes a rough pass to gauge the areas that contain the bulk of the important content (i.e., the surfaces of the objects). It then concentrates the sampling for the second stage, which involves a more refined approach. The dual sampling strategy significantly improves the accuracy and efficiency of the rendering process.

What NeRF Means for the Future of 3D Rendering

NeRF promises results that are not only new viewpoints but fully photorealistic high-resolution renderings that include complex geometry, and even real-time effects such as reflective and refractive surfaces. Older approaches get bogged down with these details and/or require extremely large voxel grids, up to 15 GB for example, where as NeRF obtains these results with only about 5 MB in network weights. NeRF consistently surpasses the performance of advanced techniques like Neural Volumes and Scene Representation Networks both in the quantitative metrics and the qualitative visuals. The mesh details in the Lego treads and in the Microphone stand are as striking as the results shown in figures 5 and 6 in the paper. Their videotaped performance is also impressive and exemplifies the smooth results of a process devoid of artifacts.

What does this mean for us?

Instantaneous 3D asset creation: Scan a real object or scene with a few photos, and boom – a perfect 3D representation ready for anything.

Unparalleled VR/AR experiences: Creating immersive virtual worlds that are indistinguishable from reality just got a whole lot closer.

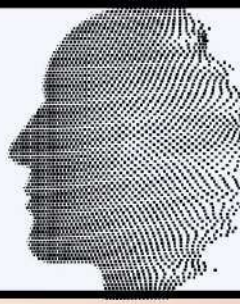
Next-gen gaming environments: Photorealistic levels without the immense manual modeling effort.

Revolutionized visual effects: Seamlessly integrating real-world scenes into films and animations.

This is an unprecedented advancement, transforming the vision of generating captivating, animated 3D visuals from ordinary images into a feasible opportunity. Neural networks can now accurately capture and learn the structure and features of our reality.

The Black Box Problem: Why We Should Understand AI Instead of Just Trusting It

Om Bansal, AIML 2024-28



As students of AIML, we all go through this. You train a large model, a huge neural network, an inverse-boosted ensemble, and with a few dozen parameters, a model achieves a solid 99% accuracy on the test set. How did the model reach this conclusion, and which features were important in that decision, what if the model made the correct prediction for all the wrong reasons, My dear reader, this is the essence of the "black box" problem in AI, which is why Explainable AI (XAI) is, and continues to be, one of the most crucial branches of Responsible AI.

The "Why" Matters More Than You Think

For a long time, the dominant mindset in ML was "performance at all costs." If the model works, who cares and how it works , But as AI starts making decisions in high-stakes domains like healthcare, finance, and criminal justice, this mindset becomes not just naive but dangerous. Imagine these scenarios:

- A deep learning model denies a loan application. The bank is legally required to provide a reason, but the model can't give one.
- A medical AI diagnoses a patient with a rare disease. The doctor, a human responsible for the patient's life, needs to understand the diagnosis to trust it and plan treatment.
- A self-driving car chooses to swerve in a critical moment. Investigators need to know why to assign responsibility and prevent future accidents.

A Toolkit for Peeking Inside the Black Box

So, how do we crack open the black box, XAI isn't a single tool, but a growing toolbox of techniques that help us understand model behavior. Let's break down a few key approaches we can actually use in our projects:

1. Model-Agnostic Methods: The Universal Interpreters

These are like diagnostic tools you can use on any model, from a random forest to a 100-layer neural net.

- **SHAP (SHapley Additive exPlanations):** This is probably the most popular tool in the XAI toolkit. Based on cooperative game theory, SHAP assigns each feature in your dataset an "importance value" for a specific prediction. It tells you exactly how much each feature (e.g., 'age', 'income') pushed the model's output from a base prediction to the final result. It's brilliant for answering: "What factors contributed most to this individual decision?"
- **LIME (Local Interpretable Model-agnostic Explanations):** LIME works by creating a simple, interpretable "surrogate model" (like a linear regression) that approximates the complex model's behavior around a single prediction. It's like zooming in on one data point and drawing a simple, understandable trend line to explain what's happening in that local region.

2. Model-Specific Methods: Built-in Transparency

Some models are designed to be more transparent from the start.

- **Attention Mechanisms:** In Transformer models (like BERT or GPT), the attention weights show you which parts of the input text the model was "paying attention to" when making its prediction. This gives a direct, intuitive view into the model's reasoning process.
- **Convolutional Neural Network (CNN) Visualization:** Techniques like Grad-CAM can generate heatmaps that highlight the regions of an image that were most influential in a classification decision. For example, it can show if a model correctly identified a "cat" by looking at its face, or incorrectly by focusing on a grassy background.

The Trade-offs: The Explainability-Performance Balance

- **High Explainability, Lower Performance:** Linear models, decision trees. We can easily understand their logic, but they may struggle with highly complex patterns.
- **High Performance, Lower Explainability:** Deep neural networks, complex ensembles. They can model incredibly nuanced relationships but are inherently opaque. The goal of XAI is not to replace all complex models with simple ones. It's to use our explainability tools to bridge this gap to let us use powerful models while still maintaining a level of understanding and control.

Why This Should Matter

1. Debugging and Improvement: Explanations help us find flaws. A model might be achieving high accuracy by "cheating" learning from a data artifact instead of the actual signal. XAI techniques can reveal this.

2. Building Trust: If you ever want a doctor, a loan officer, or a judge to use a system you built, you'll need to be able to explain it to them. Trust is the currency of real-world AI deployment.

3. Meeting Regulations: Laws like the EU's GDPR already include a "right to explanation." Understanding XAI is becoming a legal and professional necessity.

CYBERSECURITY

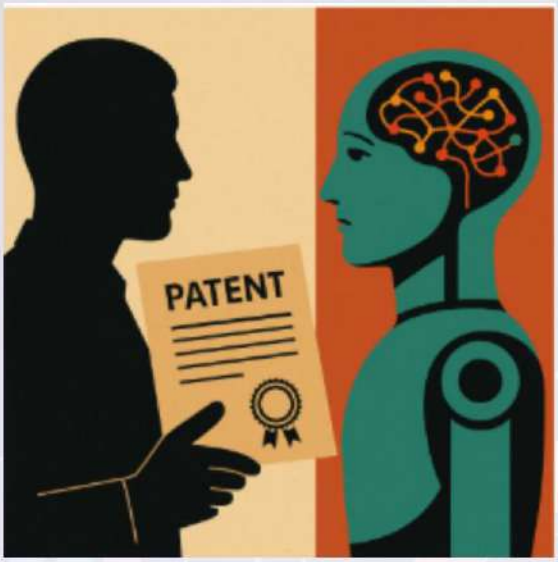
PROTECTING THE DIGITAL FRONTIER



Rethinking Creativity: AI and Intellectual Property

Prisha Vohra, AIML 2023-27

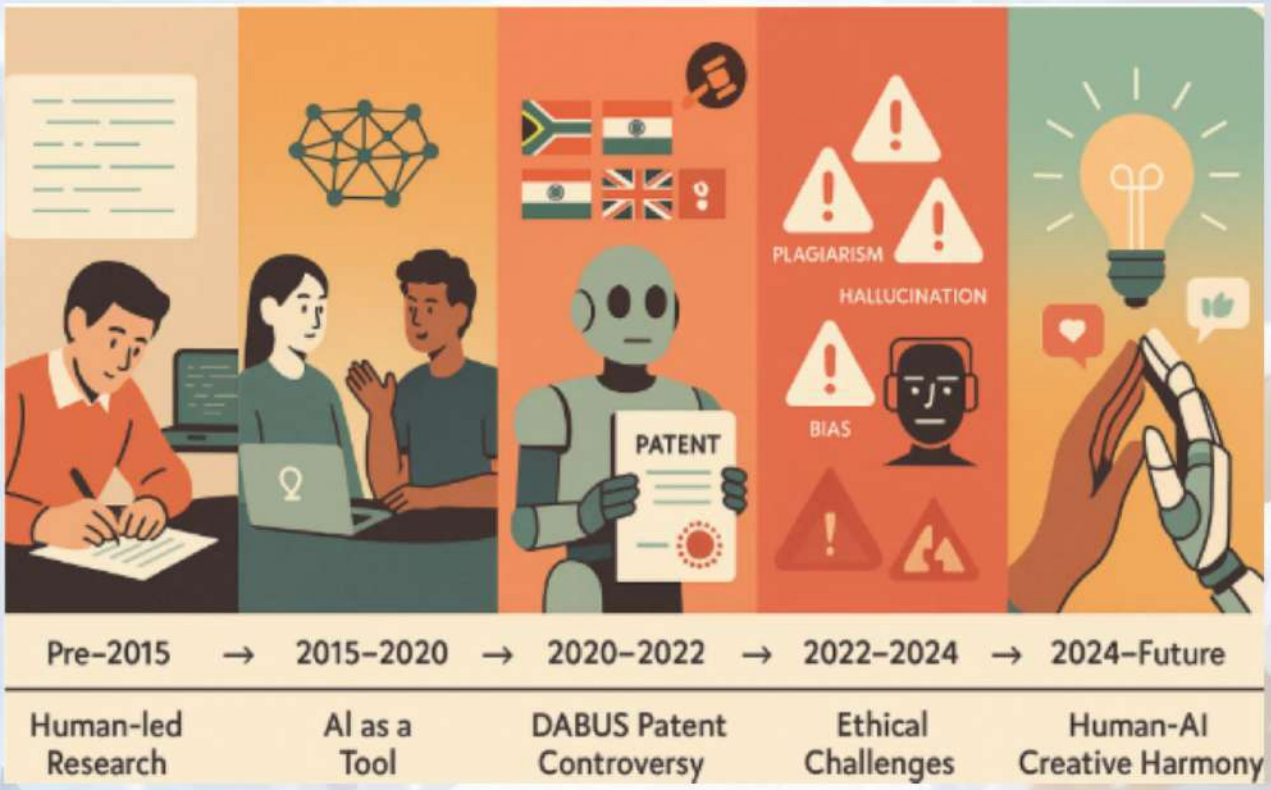
A patent application lands on the desk of an examiner. The inventor's name? Not a scientist, not an engineer—just five letters: DABUS. Not a person, but an artificial intelligence. The question arises: Can a machine truly be an inventor? This isn't a scene from a futuristic novel, it's a real dilemma facing researchers and ethicists today. As AI evolves and is utilised by humans not just to write and assist but also to brainstorm, innovate and ideate, today's AI has transitioned from a passive tool to an active creator and the line between human and machine innovation is blurring faster than ever.



Generative AI tools like ChatGPT, DALL·E, GitHub Copilot, and AlphaFold are now part of the research process in labs and universities around the world. They help write sections of papers, code, generate figures, and innovate alongside us. But does that make AI a collaborator... or does it remain just a tool? And, when an AI invents something truly new, who deserves the credit? The DABUS Dilemma: Can a Machine Be an Inventor? The story of DABUS—a neural network designed by Dr. Stephen Thaler—sent shockwaves through the intellectual property world. DABUS generated two inventive products: a novel food container and an emergency beacon. When Dr. Thaler filed for patents listing DABUS as the inventor, South Africa said "yes" — the machine deserves acclaim. The US, UK, and Europe said, "absolutely not." Their reason? AI doesn't have consciousness, intent, or legal standing. In other words, it can't own an idea. But here's the dilemma: if AI played a key role in creating something novel, does ignoring that fact make the system — and the research — less transparent? The Ethical Minefield: Who's Responsible? Leading journals like Nature and Science say AI cannot be listed as an author. Their reasoning: AI can't take responsibility, understand ethics, or answer for mistakes. Still, if AI is doing some of the intellectual heavy lifting, leaving it out of the credits entirely is intellectually dishonest. With AI's growing influence, the ethical terrain gets murky fast. If AI plagiarizes, fabricates data, or injects bias—who answers for it? The researcher, the developer or the institution? Recent controversies have shown how AI-generated images and text can slip into scientific papers without disclosure, making it harder to ensure research integrity and unambiguity.

Responsible AI: Where Do We Draw the Line?

To tackle these challenges, ethical guidelines are emerging—from UNESCO to OpenAI—but enforcement remains weak. Most academic institutions lack clear policies on how AI should (or shouldn't) be used in publishing or invention. But academia is still catching up — and without clear standards, we risk falling behind the technology we're supposed to regulate. Some suggest new categories for "AI-assisted" inventions, recognizing both human and machine contributions. Others call for mandatory disclosure of how AI was used in the research process.

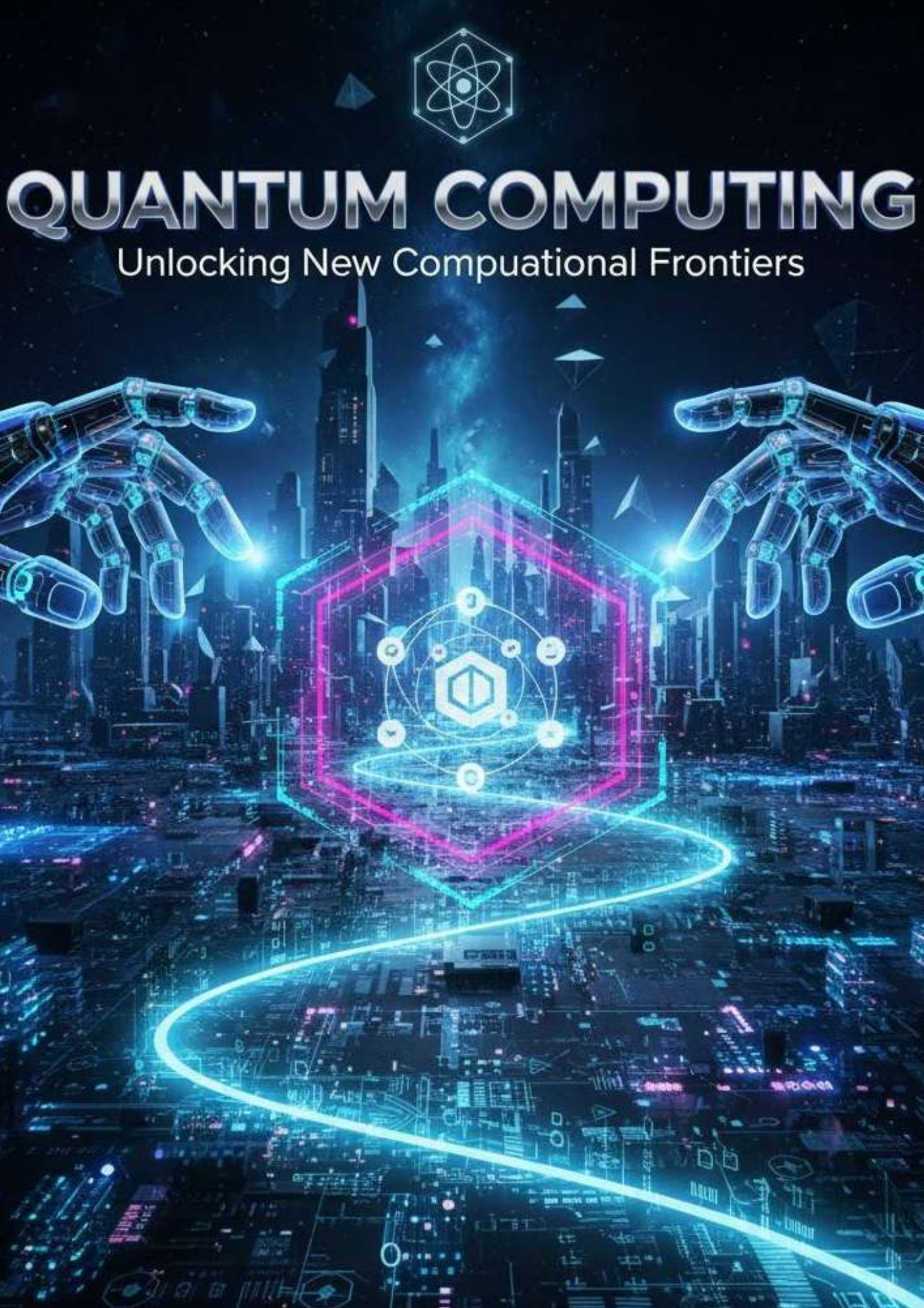


Redefining Creativity and Ownership

Perhaps the real revolution isn't about giving AI a patent or a byline, but about reimagining what it means to create and collaborate in an age of intelligent machines. The future of innovation and progress might not lie in outpacing AI, but in aligning with it—each amplifying the other's strengths. As students and future researchers, these are the questions we must confront. The choices we make now will shape the future of science, technology, and creativity itself. Because the next great inventor might not be a person —this forces us to rethink not just how we innovate, but who gets to claim innovation

QUANTUM COMPUTING

Unlocking New Compuational Frontiers





Quantum Cryptography

- **Dr. Archana Y. Chaudhari(Department of AI&ML, SIT Pune)**
- **Yashasv Khullar, AIML 2025-29**

Quantum cryptography uses the laws of quantum mechanics to achieve fundamentally secure communication. It can detect and prevent eavesdropping — even from future quantum computers.

What Is It?

Quantum cryptography applies quantum principles like superposition, entanglement, and the no-cloning theorem to ensure any interception alters the quantum state, alerting the communicating parties.



Dr Archana Y. Chaudhary
Assistant Professor,
AIML Department

Key Technique — Quantum Key Distribution (QKD)

QKD allows two users (Ram and Laxman) to share a secret key securely:

1. Ram encodes random bits on photons using different polarizations.
2. Ram sends them to Laxman through a quantum channel.
3. Ram measures with random bases.
4. They compare bases publicly and keep only matching ones (sifting).
5. They perform error correction and privacy amplification to form a final key.
6. Any eavesdropper (Eve) introduces detectable errors — if the rate is too high, they abort.

Why It's Powerful?

- Detects eavesdropping instantly
- Secure even against quantum computers
- Based on physical laws, not computational difficulty

Post-Quantum Cryptography (PQC)

While QKD uses quantum channels, PQC involves classical algorithms (lattice-, code-, hash-, or multivariate-based) designed to resist quantum attacks. It complements QKD for practical, software-based quantum-safe encryption.

Challenges

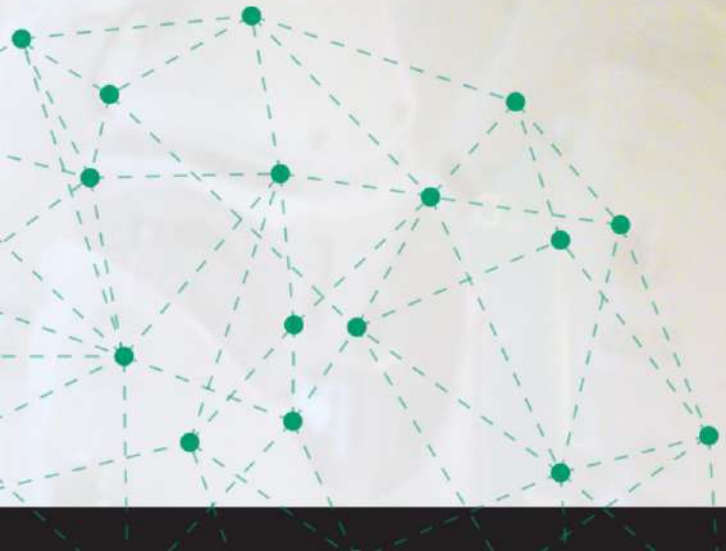
- Needs specialized quantum hardware
- Limited distance and lower key rates
- Integration with classical networks
- Vulnerable to side-channel and device imperfections

Applications

- Defence and government communication
- Banking and financial networks
- Critical infrastructure (telecom, power grids)
- Long-term confidential data protection

Future Outlook

As quantum computers mature, classical encryption (RSA, ECC) will break. The future lies in hybrid systems combining QKD + PQC for true quantum-safe security, security ensured by the laws of nature.





When Quantum Computing Meets AI

Sarvesh Sandeep Dungarwal, AIML 2024-28



You know how sometimes you're stuck on a problem and you wish you could explore every possible solution at once? Yes, that's basically what quantum computers can do and when you combine that with AI, things start to get incredible.

Quantum Machine Learning (QML) sounds like something from a sci-fi movie, but it is happening right now in big labs and companies around the globe.

Think of it this way: your laptop tries to process information bit by bit, like reading a book one word at a time. A quantum computer? It's more like absorbing an entire library simultaneously.

Why This Actually Matters

Here's where it gets interesting. Regular computers hits walls when problems become massively complex, like figuring out how proteins fold or predicting market crashes. They just cannot crunch through all the possibilities fast enough. But quantum computers use something called **qubits**, which can exist in multiple states (like 0 and 1) at once. It is like being able to take every path through a maze simultaneously instead of trying them one by one.

When researchers started combining this quantum weirdness with machine learning algorithms, something clicked. **Quantum Support Vector Machines (QSVM's)** can spot patterns in data that would make traditional algorithms throw up their hands in defeat of pursuit. **Quantum Neural Networks (QNN's)** use quantum interference, yes, actual physics, to learn faster and better.

The coolest part being that these aren't just theoretical improvements. A recent study showed that quantum systems can learn properties of other quantum systems using exponentially fewer experiments than classical methods. That's not just faster, that's fundamentally different.

Real Companies with Real Results

This is more than just academic research anymore. In 2024 and 2025, we have seen QML transition from "maybe someday" to "it is working now".

For example, take drug discovery: Companies like Google and IBM are using quantum simulations to understand how proteins fold, something that is very crucial for developing new medicines. One biotech company cut their drug development timeline by 40% using quantum-powered protein simulations. When you're talking about cancer treatments or Alzheimer's drugs, that timeline will make a difference.

JPMorgan Chase is using quantum-enhanced AI to catch fraud and run market simulations. They've cut their risk calculations from hours down to minutes. Volkswagen partnered with a quantum company and saved over $12 million annually by optimizing their robotic workflows - a 55% improvement in efficiency.

How It Actually Works

The current approach is pretty clever. Today's quantum computers are still noisy and limited, we call them NISQ devices (Noisy Intermediate-Scale Quantum). So researchers use a hybrid approach: quantum processors handle the heavy computational lifting while classical computers do the setup and optimization.

The algorithms have great names too. There's **QAOA** (Quantum Approximate Optimization Algorithm), which basically encodes your problem into quantum states and lets them evolve toward the best solution. And **VQE** (Variational Quantum Eigensolver) combines quantum preparation with classical optimization, it is like having the best of both worlds.

Some researchers are getting 10-million-fold speedups with **Quantum Tensor Networks**. Others have shown that you can get better machine learning results using just two photons than with classical methods.

The Challenges (Because There Always Are Some)

Let's be honest, this stuff is not easy. Current quantum computers are finicky. They're sensitive to noise, don't have many qubits and lose their quantum properties quickly. It's like trying to do precise calculations while someone is shaking your desk.

Scaling up is tricky too. Researchers are getting creative with things like multi-chip approaches, where they split computations across several smaller quantum devices. And finding people who understand both quantum physics and machine learning? That's a challenge in itself.

Why You Should Care

Here's the thing - this field is exploding. Over $1.2 billion got invested in quantum startups just in the first quarter of 2025. Universities are rolling out quantum machine learning courses, and companies like IBM offer free quantum programming platforms. NASA, Los Alamos and other major labs are offering research internships.

Whether you're into physics, computer science, math or engineering, there's a way in. Current research covers everything from quantum natural language processing to optimization algorithms that could revolutionize logistics.

Google's latest quantum processor, Willow, just did calculations in minutes that would take classical computers billions of years. We're not just talking about incremental improvements; we are looking at fundamentally new ways of solving problems.

The quantum revolution in AI is not coming someday. It is happening now and there's room for curious minds like yours to help shape where it goes next.

The Future Is Intelligent — and So Are You.

As we turn the last page of this edition, we're reminded that AI is not just about algorithms — it's about ideas that learn, adapt, and inspire.

From neural networks that mimic the brain to ethical frameworks that guide innovation, every article, project, and insight in this magazine represents one truth:

The future belongs to those who can think both logically and responsibly.
At the intersection of technology and humanity, Artificial Intelligence continues to redefine what's possible — in classrooms, industries, and imagination.

Whether you're a researcher, student, or enthusiast, remember —

- Keep questioning.
- Keep experimenting.
- Keep learning.

Because in AI and life alike — evolution never stops.

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